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The Norwegian Petrocurrency Puzzle

Oil, Fundamentals, and Financial Conditions: Determinants of the Norwegian Krone in a Multi-Country Framework

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Abstract

This thesis examines whether oil prices are a systematic and distinct driver of the Norwegian krone. The analysis addresses whether oil adds explanatory power beyond standard macroeconomic fundamentals, whether the krone responds differently from other commodity-exposed currencies, and whether these relationships vary across volatility regimes. The analysis uses a quarterly multi-country panel of fourteen economies covering 2000Q1 to 2025Q4, with a sequence of nested panel specifications featuring two-way fixed effects, group-level oil interactions, and volatility regimes captured by a binary high-VIX indicator. Oil prices contribute meaningful incremental explanatory power beyond the macroeconomic baseline. Higher oil prices are associated with an appreciation of the Norwegian krone. This relationship is statistically robust, stronger than that of the diversified reference, and not distinguishable from that of other oil-exporting currencies when averaged over the full sample. The relationship varies systematically with the volatility regime, with non-Norway commodity groups acquiring substantial oil sensitivity primarily under elevated volatility, while Norway maintains a response of comparable magnitude across regimes. This regime stability distinguishes the krone within the commodity-currency class. The thesis sharpens the oil-currency claim for the Norwegian krone by showing that the krone's distinctness emerges in calm market conditions and narrows under stress as other commodity-exposed currencies converge toward Norway's magnitude.

Keywords - Exchange rates, petrocurrency, Norwegian krone, oil prices, state dependence

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1 Introduction

1.1 Motivation and background

The relationship between oil prices and exchange rates has long played a central role in both academic research and policy discussions. For countries with substantial commodity production, movements in global commodity prices are often believed to exert a strong influence on the value of the domestic currency. The Norwegian Krone, as the currency of a major exporter of oil and gas, is frequently cited as a prototypical “oil currency,” since movements in oil prices are assumed to translate directly into movements in the Norwegian krone.

Empirical research has documented persistent relationships between oil prices and exchange rates. Amano and van Norden (1998a) identify a stable long-run relationship between oil prices and the U.S. real exchange rate, with the dollar appreciating when oil prices rise. A natural interpretation is that oil is traded in U.S. dollars, which generates additional demand for dollars from importing countries when oil prices rise. Yet exchange rates of other currencies, including the Norwegian krone, are also frequently described as co-moving with oil prices. Since the dollar mechanism applies directly only to the dollar, this co-movement must reflect channels operating beyond the global pricing role of the dollar.

The perception of the Norwegian krone as an “oil currency” is widely reflected in market commentary and institutional analysis. Norges Bank and market participants frequently point to oil prices as an important factor in explaining movements in the Norwegian krone, particularly during periods of large commodity-price fluctuations (see Norges Bank, 2022, 2025, 2026; ter Ellen, 2016; ter Ellen & Martinsen, 2016). At the same time, Norway’s macroeconomic framework has evolved substantially over the past decades. The continued expansion of the Government Pension Fund Global and shifts in monetary policy regimes may have altered the channels through which oil prices affect the exchange rate. Despite this widespread perception, the empirical relationship between oil prices and the Norwegian krone has long been more complex than commonly assumed. Akram (2004) finds the relationship to be non-linear, with the strength of co-movement varying systematically across different oil-price environments.

1.2 Research question

The academic literature provides mixed evidence on this issue. Some studies find that the predictive content of oil and commodity prices for exchange rates is confined to very high frequencies, with little systematic relation at monthly or quarterly horizons (Ferraro et al., 2015). At the same time, more recent research shows that the evidence on the oil-exchange rate relationship varies substantially depending on sample, country choice, and empirical method, and that the effects are strongly time-varying (Beckmann et al., 2020). Recent structural analyses, including Bjørnland et al. (2024), suggest that low-frequency movements in real exchange rates can be linked to productivity shifts and commodity market trends, including oil prices, as part of broader macroeconomic dynamics.

Taken together, these findings imply that oil prices may act both as long-run fundamentals and as short-run financial signals, but that their explanatory role is neither universal nor constant. This raises a fundamental question in the Norwegian context:

To what extent are oil prices a systematic and distinct driver of the Norwegian krone, and under what conditions does this relationship hold?

This overarching question can be divided into three more specific empirical questions:

1. Do oil prices add explanatory power for exchange-rate movements once standard macroeconomic fundamentals are taken into account?

2. Does the Norwegian krone exhibit greater oil sensitivity than other commodity-exporting economies, and does this reflect something distinct about Norway specifically?

3. Does the relationship between oil prices and exchange rates vary across volatility regimes, such that oil sensitivity is stronger or weaker during periods of financial stress?

1.3 Conceptual perspective

From a behavioural finance perspective, this question is directly relevant. Exchange rates are not only determined by macroeconomic fundamentals but are also financial asset prices, traded continuously by participants whose expectations, risk perceptions, and information processing capacity vary over time. In calm environments, pricing may largely reflect underlying economic conditions. In periods of elevated uncertainty, market participants may rely more heavily on salient narratives and simplified characterizations of currencies, a pattern consistent with Shiller's (2017) argument that economic narratives can spread and influence asset prices independently of underlying fundamentals. The characterization of the Norwegian krone as an "oil currency" can be understood in this light. It may function as a reflection of the underlying economic structure, but also as a persistent market narrative that gains influence when uncertainty is high and simpler signals become more attractive to market participants. This motivates an empirical design that allows the role of oil to vary with financial market conditions, so that any systematic deviations from fundamental drivers can be identified and interpreted considering the behavioural perspective outlined above.

1.4 Empirical approach

The purpose of this thesis is to re-examine the role of oil prices in exchange rate determination within a broader macroeconomic and financial framework. Rather than treating oil prices as an inherent driver of the Norwegian krone, the analysis assesses whether oil adds explanatory power once exchange rate movements are evaluated alongside standard macroeconomic fundamentals.

The empirical strategy proceeds through a sequence of model specifications. We begin with a macroeconomic baseline and subsequently extend the framework to incorporate oil price exposure, financial market conditions, and potential state dependence. In particular, we allow the impact of oil prices to vary across market environments, enabling an assessment of whether oil-related exchange rate effects intensify during periods of heightened uncertainty. The analysis is conducted within a multi-country setting at quarterly frequency, a choice motivated by what Obstfeld and Rogoff (2001) call the exchange-rate disconnect puzzle, relating to the weak short-term feedback links between exchange rates and fundamentals. The quarterly frequency adopted here abstracts from these short-run dynamics and focuses attention on the slower fundamental drivers of the exchange rate, an approach consistent with Bjørnland et al. (2024).

1.5 Contribution

This thesis contributes to the literature in three related ways. First, it reassesses the role of oil in exchange-rate determination within a disciplined benchmark framework. By beginning with standard macroeconomic controls and then asking whether oil adds explanatory power beyond them, the thesis speaks directly to the question of whether the oil-currency narrative retains incremental empirical relevance.

Second, the comparative panel design isolates oil sensitivity as a dynamic response rather than a structural characteristic. Country fixed effects absorb time-invariant differences between countries, including their average level of oil exposure, so the analysis tests whether the krone responds systematically more to quarterly oil-price changes than other commodity-exposed currencies after such differences are accounted for.

Third, the design allows the comparative oil response of each currency group to vary between calm and stressed market conditions, capturing state dependence in the oil-exchange rate relationship that single-currency or unconditional specifications cannot detect.

More broadly, the thesis does not claim to overturn or replace existing theory of exchange-rate determination. Its contribution is instead to sharpen a familiar claim by asking whether the oil sensitivity of the Norwegian krone is better understood as conditional, comparative, and regime-dependent rather than as a fixed structural feature of the currency.

1.6 Limitations

The thesis is deliberately reduced form in its empirical design. The results should be interpreted as evidence on the comparative and state-dependent sensitivity of exchange rates to oil prices, not as identification of a single structural causal channel. In addition, the use of quarterly data implies that the analysis focuses on the slower, macro-driven relationships that operate at quarterly frequency rather than on short-run, high-frequency exchange rate dynamics, and should be interpreted accordingly.

1.7 Structure of the thesis

The remainder of the thesis is structured as follows. Section 2 reviews the theoretical and empirical literature on exchange rate determination, commodity price channels, financial integration, and behavioural perspectives. Section 3 develops the empirical framework and outlines the baseline and extended model specifications. Section 4 describes the data, variable construction, and frequency choices. Section 5 presents the econometric methodology and identification strategy. Section 6 reports the main empirical results. Section 7 provides robustness analyses and alternative specifications. Section 8 discusses the findings against the theoretical framework, and Section 9 concludes.

2 Theoretical Framework and Literature Review

This chapter reviews the theoretical and empirical literature relevant to the thesis. The chapter begins by outlining the asset-pricing view of exchange rates and the role of standard macroeconomic benchmark variables. It then turns to the commodity-currency literature and the channels through which commodity prices may affect exchange rates. Finally, it discusses the role of global risk and volatility regimes in shaping exchange rate dynamics.

2.1 Exchange rates as asset prices and benchmark models

Exchange rate theory treats exchange rates as forward-looking asset prices rather than as variables determined solely by contemporaneous trade flows or relative goods prices. In this perspective, the value of a currency reflects current macroeconomic conditions, as well as expectations about future fundamentals, monetary policy, and the compensation investors require for holding country-specific risk (Obstfeld & Rogoff, 1996). Importantly, this implies that exchange rates may at times move in response to shifts in expectations, risk appetite, or market narratives even when underlying fundamentals have changed little.

This broad perspective encompasses several potential sources of exchange rate variation, of which macroeconomic fundamentals are only one. Yet much of the empirical literature has focused on whether observable macro variables alone can account for currency movements. Purchasing power parity, or PPP, provides a long-run benchmark according to which nominal exchange rates should adjust to offset relative price-level differences across countries.

Empirically, however, deviations from PPP can persist for extended periods, particularly at short and medium horizons (Rogoff, 1996). Uncovered interest parity, or UIP, links expected exchange rate changes to short-term interest rate differentials. Yet UIP also performs poorly in the empirical data. Fama (1984) demonstrates that the forward premium is a biased predictor of future exchange rate changes, suggesting that time-varying risk premia play a significant role in currency markets. More recently, Baillie et al. (2026), while acknowledging the broad empirical evidence against UIP with monthly data, find that the parity does hold in the long run. More broadly, Meese and Rogoff (1983) show that standard macroeconomic exchange-rate models struggle to outperform a random walk in out-of-sample forecasting exercises.

These findings highlight a persistent gap between what macroeconomic fundamentals can explain and how exchange rates actually behave. This gap has motivated a broader literature that examines whether additional factors, such as commodity prices, financial conditions, and volatility regimes, can account for the unexplained variation in exchange rate movements.

2.2 Macroeconomic fundamentals and parity relations

Despite the empirical limitations documented in Section 2.1, parity conditions remain the theoretical foundation of exchange rate economics. The most direct link is provided by uncovered interest parity. As a no-arbitrage condition, UIP requires that the expected return on domestic and foreign assets is equalized when expressed in a common currency:

$$\frac{E_t[S_{t+1}]}{S_t} = \frac{1 + i_t}{1 + i_t^*} \tag{2.1}$$

where S_t is the nominal exchange rate (domestic currency per unit of foreign currency), i_t the domestic short-term interest rate, and i_t^* the foreign equivalent. Taking logarithms and applying $\ln(1 + x) \approx x$:

$$E_t[s_{t+1}] - s_t \approx i_t - i_t^* \tag{2.2}$$

where lowercase letters denote logarithms. Rearranging and defining the interest rate differential $i_t^D = i_t - i_t^*$, the current exchange rate can be expressed as $s_t = E_t[s_{t+1}] - i_t^D$.

Substituting forward recursively under the law of iterated expectations and letting $\mu = lim_T \rightarrow \infty, E_t[s_{t+T}]$ denote the expected long-run exchange rate:

$$s_t = \mu - \sum_{j=0}^{\infty} E_t\left[i_{t+j}^D\right] \tag{2.3}$$

The spot rate today thus equals the expected long-run rate minus the entire expected path of future interest rate differentials. Because the summation extends to infinity, even small revisions in distant expectations can generate large movements in the current rate. Engel and West (2005) show that this property can produce near-random-walk behaviour even when fundamentals genuinely drive the exchange rate, providing a structural interpretation of the Meese and Rogoff (1983) result. In practice, the UIP condition is violated systematically, as documented in Section 2.1. The empirical failure of UIP implies that exchange rate movements reflect not only interest rate differentials but also time-varying risk premia and other factors that lie outside the scope of standard parity conditions.

While UIP concerns arbitrage in financial markets, purchasing power parity rests on arbitrage in goods markets, implying that exchange rates should adjust over time to offset differences in national price levels. Formally:

$$S_t = \frac{P_t}{P_t^*} \text{ or equivalently } s_t = p_t - p_t^* \tag{2.4}$$

where P_t and P_t^* are the domestic and foreign price levels. These parity conditions thus play a dual role. They establish that interest rate differentials and relative price levels form the theoretical core of exchange rate determination, and any credible empirical framework must account for them. At the same time, their persistent empirical failure confirms that these forces alone leave a substantial share of exchange rate variation unexplained. For Norway specifically, Bjørnland and Hungnes (2002), modelling the Norwegian exchange rate against a basket of trading partners, find a robust long-run link between the real exchange rate and the real interest differential that is consistent with PPP and UIP holding jointly, while PPP alone is rejected.

2.3 Commodity currencies and oil price transmission channels

The gap between fundamentals-based predictions and observed exchange rate behaviour, documented in the previous section, has motivated a parallel literature that looks beyond parity conditions. The commodity currency strand of this literature emphasizes several distinct channels through which commodity prices may influence exchange rates, the relative importance of which depends on a country's economic structure, degree of commodity exposure, and institutional framework. Chen and Rogoff (2003) provide the foundational contribution, showing that world commodity prices exhibit a strong and stable empirical relationship with the real exchange rates of small commodity-exporting economies, where commodity export prices are generally treated as exogenous. This motivates the treatment of commodity prices as a potential explanatory factor in exchange rate determination, an approach that has since been extended and qualified by a growing body of empirical work.

The most direct channel operates through the terms of trade. For a commodity-exporting economy, an increase in global commodity prices raises the value of exports relative to imports. As export revenues are converted into domestic currency to meet domestic operating costs, wages, and tax obligations, the demand for the domestic currency strengthens and puts upward pressure on the exchange rate. Amano and van Norden (1998b) document this mechanism by linking oil prices to the real exchange rate through their effect on terms of trade, and Beckmann et al. (2020) identify it as the primary direct transmission channel in their survey of the oil price and exchange rate relationship.

A second direct channel operates through wealth effects and portfolio rebalancing. Krugman (1983) and Golub (1983) develop frameworks in which higher oil prices transfer wealth from oil-importing to oil-exporting economies, with the resulting exchange rate adjustments arising from differential portfolio preferences across countries. The strength of this channel depends on how oil revenues are absorbed into the economy. When revenues are largely reinvested in foreign-currency assets, appreciation pressures on the domestic currency are dampened. When revenues are spent or invested domestically, appreciation pressures intensify. The channel therefore varies across commodity-exporting countries.

Beyond the direct terms-of-trade mechanism, the literature typically treats oil prices as closely linked to inflation, real activity, and the monetary policy response to them. Structural analyses such as Kilian and Zhou (2022) isolate individual shocks to the oil market, the dollar, and the U.S. real interest rate and document that these variables are determined simultaneously. In a panel of thirty economies, Attílio and Mollick (2026) find that interest rates in commodity-exporting countries including Canada, Norway, and South Africa move positively in response to oil price shocks. The strength of the channel is not uniform, however. Kilian and Lewis (2011) find no credible evidence that the Federal Reserve has systematically responded to oil price shocks over their sample. Whether oil prices continue to matter for the exchange rate once observed interest rate differentials are taken into account is therefore an empirical question, and one whose answer may well depend on the country and period under consideration.

A separate strand of the commodity currency literature studies the long-run structural consequences of sustained resource income, most prominently within the Dutch disease framework introduced by Corden and Neary (1982). The model identifies two distinct effects. The resource movement effect reflects that labour and capital flow toward the booming resource sector, reducing inputs available to other tradable industries. The spending effect describes how resource income is spent within the economy, raising demand for non-tradable goods and services. Both effects push up the relative price of non-traded to traded goods. Since non-tradable prices are not disciplined by international arbitrage, this translates into real exchange rate appreciation.

A related but distinct long-run mechanism is the Balassa-Samuelson effect (Balassa, 1964; Samuelson, 1964), which operates through the supply side. Faster productivity growth in the tradable sector drives up wages across the economy, and since productivity in the non-tradable sector does not keep pace, prices there must rise to cover the higher wages, producing real appreciation through the same relative price channel. In resource-rich economies, the two mechanisms can be active simultaneously, since the commodity sector is typically both a major source of aggregate income and highly productive per worker. While the terms-of-trade and monetary policy channels explain how the exchange rate reacts to movements in the oil price, these structural channels describe the long-run backdrop against which these reactions play out. In economies with a large and persistent commodity sector, these structural channels add to the set of mechanisms through which commodity prices may influence exchange rate behaviour.

Taken together, the commodity currency literature points to a layered set of mechanisms through which oil prices may enter exchange rate dynamics in commodity-exposed economies, but their empirical relevance varies across countries and periods. The next section turns to the broader financial environment and its role in shaping exchange rate dynamics.

2.4 Financial conditions, risk premia, and global factors

Beyond macroeconomic fundamentals and commodity-related channels, exchange rate dynamics are also shaped by global financial conditions. Shifts in global risk appetite, capital flows, and market-wide volatility influence how currencies trade, and the very relationships between fundamentals and exchange rates can vary with these conditions. This section turns to the literature on these factors, with particular attention to how volatility regimes shape exchange rate behaviour.

A core finding in this literature is that exchange rates co-move systematically with global financial conditions, and that the strength of this co-movement varies across currencies. Lustig et al. (2011) identify a common global risk factor that explains most of the variation in excess currency returns across countries, with high-interest rate currencies more exposed to this factor than low interest rate currencies. They document that this factor is related to changes in global equity market volatility, pointing to external global risk as a driver of currency returns. Menkhoff et al. (2012) offer an alternative perspective, arguing instead that volatility within the foreign exchange market itself is the systematic risk factor that prices currency returns, with high-yield currencies delivering low returns in times of unexpectedly high FX volatility.

This literature further documents that the relationship between global financial conditions and exchange rates is particularly pronounced during periods of market stress. The carry trade provides a well-known example. Carry strategies, which exploit persistent violations of uncovered interest parity by borrowing in low-yield currencies and investing in high-yield currencies, typically generate steady returns during calm periods but are exposed to abrupt reversals when global risk rises. Brunnermeier et al. (2008) document that these reversals coincide with sharp increases in the VIX and with contractions in funding liquidity. The pattern is often summarized by the adage that carry trades “go up the stairs and come down the elevator”, capturing the asymmetric dynamics in which risk-sensitive currency relationships gradually build up and unwind abruptly.

Beyond these high-frequency carry dynamics, a broader literature establishes that VIX-based financial conditions shape international financial outcomes at lower frequencies. Forbes and Warnock (2012), using quarterly data for 58 countries, find that global risk is the most consistently significant predictor of extreme capital flow episodes, including sudden stops in foreign inflows and retrenchments by domestic investors. Rey (2015) using a recursive VAR on quarterly data, shows that VIX shocks produce persistent responses in cross-border capital flows, bank leverage, and global credit over horizons of four to fifteen quarters.

A closely related concept in this literature is that of safe haven currencies, which tend to appreciate during periods of global uncertainty and thus provide a hedge against risky asset portfolios. Ranaldo and Söderlind (2010) document that the Swiss franc and the Japanese yen strengthen against the US dollar when US stock prices fall and FX volatility rises, with these safe haven effects particularly pronounced during crises. Habib and Stracca (2011) extend this perspective by analysing a large panel of currencies and find that safe haven status is associated with structural characteristics such as net foreign asset positions and the size of the domestic financial market, rather than with interest rate levels alone. These findings suggest that the direction of a currency's response to global risk depends on structural features that distinguish safe haven currencies from those exposed to commodity cycles or other risk factors, and that safe haven and commodity currencies may respond systematically differently to the same global risk episode.

Taken together, this literature highlights that exchange rate dynamics cannot be understood in isolation from global financial conditions. The strength of this relationship varies across currencies, across time, and with the structural features of the issuing economy.

2.5 Synthesis of the literature

The literature reviewed in the preceding sections points to three broad insights. Exchange rates respond to a range of macroeconomic fundamentals, but standard models have limited empirical success, particularly at short horizons, and systematic deviations from uncovered interest parity suggest that risk premia and other factors outside the standard framework play an important role. Commodity prices appear relevant for exchange rate dynamics through several identifiable channels, especially for resource-rich economies, but the observed relationships are not uniform and vary with country characteristics, data frequency, and time period. Global financial conditions shape exchange rate behaviour on their own and may also alter the strength of other relationships, with risk exposure differing across currencies and across market environments.

These three strands are not independent. A full account of exchange rate dynamics in commodity-exposed economies requires considering macroeconomic fundamentals, commodity-related channels, and global financial conditions jointly, since the empirical relevance of each may depend on the others.

3 Empirical Framework and Model Design

This chapter develops the empirical framework used to evaluate the research question set out in Chapter 1. The purpose is to translate the theoretical discussion in Chapter 2 into a sequence of estimable specifications that address three questions: whether oil prices add explanatory power for exchange-rate movements once standard macroeconomic fundamentals are controlled for, whether this oil sensitivity differs across commodity-exposed currencies, and whether it varies across volatility regimes. The chapter presents the structure of the panel, the baseline specification, and the extensions used to address each research question.

3.1 From theory to testable specifications

The empirical analysis is conducted as a panel study of quarterly exchange-rate changes. As discussed in Section 1.4, the quarterly frequency is chosen to focus on the slower fundamental drivers of exchange rates. The panel includes fourteen economies organised into four groups, with Norway treated separately, and the United States serving as the common numeraire. Estimation is therefore performed on fourteen bilateral exchange-rate series against the U.S. dollar. A panel design is well suited to the research question because it combines time-series variation and cross-country variation in a single framework, allowing us to analyse whether the krone responds to oil differently from other commodity-linked currencies, and whether this relative response varies across market environments.

The panel framework includes both country fixed effects and time fixed effects. Country fixed effects absorb time-invariant structural differences across economies, including monetary-policy regime, institutional quality, trade structure, and the degree of commodity dependence. Time fixed effects absorb all shocks common to the panel in a given quarter, including global dollar cycles, changes in international risk sentiment, and major geopolitical or financial disturbances. This two-way fixed-effects structure is central to the identification strategy because the global oil price takes a common value for all panel units in a given quarter. A stand-alone oil term is therefore absorbed by the time fixed effects and cannot be estimated separately. Any oil effect must instead be identified from heterogeneous exposure across currencies.

The model does not attempt to separate underlying oil-market shocks into structural supply, aggregate demand, and oil-specific demand components, as in Kilian (2009) and Kilian and Zhou (2022). Oil-price changes enter as a summary of oil-market movements rather than as identified structural disturbances.

The empirical strategy is built around four research hypotheses, examined through a structured sequence of specifications:

Hypothesis 1: *Oil prices provide incremental explanatory power for exchange-rate movements beyond standard macroeconomic fundamentals.*

Hypothesis 2: *Oil commodity currencies exhibit greater oil sensitivity than other commodity and control currencies.*

Hypothesis 3: *The oil sensitivity of the Norwegian krone exceeds that of other oil-exporting currencies.*

Hypothesis 4: *The exchange-rate effect of oil prices varies across volatility regimes.*

The four hypotheses above are the claims this thesis sets out to test. Each is tested by evaluating the corresponding null hypothesis, the opposite of the research hypothesis. If the observed data are sufficiently unlikely under the null, it is rejected as evidence consistent with the research hypothesis.

The empirical analysis builds on a sequence of four specifications. Equation (3.1) is a country-by-country auxiliary regression that motivates the panel design. The remaining specifications form a sequence of three nested panel models. Equation (3.2) establishes a macroeconomic benchmark, equation (3.3) augments the benchmark with heterogeneous oil exposure, and equation (3.4), the preferred specification, introduces state dependence through interactions with the volatility regime.

3.2 Auxiliary time-series specification

The first specification is a simple country-by-country baseline regression that estimates a separate oil-price coefficient for each currency. For each country i, the model is:

$$\Delta s_{it} = \alpha_i + \beta_1(i_{it} - i_{US,t}) + \beta_2(\pi_{it} - \pi_{US,t}) + \beta_3(g_{it} - g_{US,t}) + d_i \Delta oil_t + \varepsilon_{it} \quad (3.1)$$

where Δs_{it} is the quarterly change in the log nominal bilateral exchange rate, $(i_{it} - i_{US,t})$, $(\pi_{it} - \pi_{US,t})$ and $(g_{it} - g_{US,t})$ are the short-term interest-rate, inflation, and GDP-growth differentials relative to the United States. Δoil_t is the quarterly change in the log oil price, α_i is a country-specific intercept, d_i is the country-specific oil-price coefficient, and ε_{it} is the error term.

The country-specific coefficients are aggregated using the Mean Group procedure of Pesaran and Smith (1995), which provides an average oil-price coefficient across the panel. This specification serves as the starting point of the empirical analysis. The panel specifications introduced in the following sections build on this baseline by exploiting the cross-country dimension more directly.

3.3 Model 1: Baseline Macro Fundamentals

The first panel specification is a baseline model of exchange-rate changes based on the macroeconomic fundamentals introduced in Section 3.2. Its role is to provide a benchmark against which any oil-related term must justify its inclusion. The baseline model is written as:

$$\Delta s_{it} = \alpha_i + \lambda_t + \beta_1(i_{it} - i_{US,t}) + \beta_2(\pi_{it} - \pi_{US,t}) + \beta_3(g_{it} - g_{US,t}) + \varepsilon_{it} \quad (3.2)$$

where α_i denotes country fixed effects and λ_t denotes time fixed effects. The coefficients β_1, β_2 and β_3 capture the average partial effect of each macroeconomic differential on exchange-rate changes, holding country and time fixed effects constant.

For compactness, the three macroeconomic differentials are collected in a vector X_{it}, and the subsequent specifications build on this compact form. The model can equivalently be written as:

$$\Delta s_{it} = \alpha_i + \lambda_t + \beta' X_{it} + \varepsilon_{it} \tag{3.2a}$$

3.4 Model 2: Commodity augmentation and oil exposure

Model 2 extends the baseline model by allowing oil to enter through interactions with country-group indicators. Unlike the macroeconomic differentials, the oil price takes the same value for all countries in a given quarter and is therefore fully absorbed by time fixed effects. Identification of oil effects must come from differences in how commodity-linked currencies respond to the same common oil-price movement. The specification distinguishes between four non-reference groups, allowing each to have its own oil sensitivity.

Let $G \in \{N, O, C, S\}$, where N = Norway, O = Non-Norwegian-Oil-Currencies, C = Other Commodity Currencies, and S = Safe Havens. The non-simplified specification is:

$$\Delta s_{it} = \alpha_i + \lambda_t + \beta' X_{it} + \theta_N(\Delta oil_t \times N_i) + \theta_O(\Delta oil_t \times O_i) + \theta_C(\Delta oil_t \times C_i) + \theta_S(\Delta oil_t \times S_i) + \varepsilon_{it} \tag{3.3}$$

Which can be written:

$$\Delta s_{it} = \alpha_i + \lambda_t + \beta' X_{it} + \sum_{g=1}^{G} \theta_g(\Delta oil_t \times Group_i) + \varepsilon_{it} \tag{3.3a}$$

The coefficient θ_N captures the oil sensitivity of the Norwegian krone. θ_O captures the corresponding sensitivity for the four other oil exporters. θ_C captures the sensitivity for the four non-oil commodity exporters, and θ_S captures the sensitivity for the two safe-haven currencies. All four coefficients are measured relative to the omitted diversified developed-market reference.

3.5 Model 3: State dependence and financial conditions

Model 2 imposes that oil sensitivity is constant across market environments. The discussion in Section 2.4 suggests that this may be too restrictive. Model 3 therefore allows oil sensitivity to differ between volatility regimes.

The $HighVIX$ indicator is used as a regime indicator that captures the overall market environment within each quarter rather than individual short-lived volatility spikes. A quarter classified as high volatility may reflect either persistently elevated stress or repeated episodes within the quarter, both of which represent an environment in which uncertainty and risk aversion are elevated.

Because $HighVIX_t$ is common across countries, it is absorbed by the time fixed effects unless interacted with country-group indicators. A direct interaction between Δoil_t and $HighVIX_t$ is likewise absorbed for the same reason. State dependence must therefore be identified through higher-order interactions that allow the comparative oil response of each commodity group to differ between volatility regimes. The model also includes lower-order interactions between the commodity-group indicators and the volatility regime, ensuring that the triple oil-by-group-by-regime interactions isolate oil-specific regime dependence rather than generic differences in how each group responds to stress periods.

$$\begin{aligned}\Delta s_{it} = \alpha_i + \lambda_t + \beta' X_{it} \\ + \sum_{g=1}^{G} \gamma_g \left(Group_{g,i} \times HighVIX_t\right) \\ + \sum_{g=1}^{G} \theta_g \left(Group_{g,i} \times \Delta oil_t\right) \\ + \sum_{g=1}^{G} \varphi_g \left(Group_{g,i} \times \Delta oil_t \times HighVIX_t\right) + \varepsilon_{it}\end{aligned} \tag{3.4}$$

The lower-order γ_g coefficients absorb generic differences in how each group responds to high-volatility periods, independent of oil-price movements and the φ_g triple interactions then identify the regime-induced shift in oil sensitivity that is specific to each group. The implied total effects for each combination are summarised in Table 3.1.

Table 3.1: Implied total oil effects for each country group in eq. 3.4

Regime	Norway	Other oil	Other commodity	Safe Haven
Low-volatility quarters	θ_N	θ_O	θ_C	θ_S
High-volatility quarters	$\theta_N + \varphi_N$	$\theta_O + \varphi_O$	$\theta_C + \varphi_C$	$\theta_S + \varphi_S$

Note. The omitted diversified developed-market reference (EUR, GBP, SEK) has no oil-specific interaction term and an implied oil sensitivity of zero by construction.

The four specifications introduced form the empirical core of the thesis. Variable construction is described in the next chapter, while estimation and inference are set out in Chapter 5.

4 Data and Variable Construction

This chapter describes the data used to estimate the specifications introduced in Chapter 3. The chapter begins by outlining the country sample and time period, then presents the exchange-rate data and the macroeconomic and financial variables that enter the models. The final section reports descriptive statistics and stylized facts that motivate the empirical analysis that follows. Table 4.1 summarises the variables used throughout the chapter:

Table 4.1: Variable definition

Symbol	Definition	Unit
Δs_{it}	Change in the log nominal bilateral exchange rate	Log points
$i_{it} - i_{US,t}$	Short-term interest-rate differential relative to the US	Percentage Points
$\pi_{it} - \pi_{US,t}$	Inflation differential relative to the US	Log points
$g_{it} - g_{US,t}$	GDP-growth differential relative to the US	Log points
X_{it}	Inflation, Interest-rate and GDP vector	N/A
Δoil_t	Quarterly change in the log Brent oil price	Log points
C_i	1 if non-oil commodity linked economy	Binary
N_i	1 if Norway	Binary
S_i	1 if Safe Haven	Binary
O_i	1 if oil linked economy other than Norway	Binary
$HighVIX_t$	1 if in high-volatility quarters	Binary

4.1 Country sample

The panel comprises fourteen economies over the period 2000Q1 to 2025Q4, organised into four groups according to their commodity exposure and exchange-rate characteristics, with Norway treated separately in the empirical analysis. The grouping reflects both the theoretical channels reviewed in Chapter 2 and the identification logic of the empirical design in Chapter 3. Table 4.2 provides an overview of the country sample and group classification.

Table 4.2: Country sample and group classification.

Code	Country	Group	Sample period	T
NOK	Norway	Norway	2000Q1–2025Q4	103
CAD	Canada	Oil commodity	2000Q1–2025Q4	103
MXN	Mexico	Oil commodity	2000Q1–2025Q4	103
BRL	Brazil	Oil commodity	2000Q1–2025Q4	103
COP	Colombia	Oil commodity	2000Q1–2025Q4	103
AUD	Australia	Other commodity	2000Q1–2025Q4	103
NZD	New Zealand	Other commodity	2000Q1–2025Q4	103
CLP	Chile	Other commodity	2001Q1–2025Q4	97
ZAR	South Africa	Other commodity	2000Q1–2025Q4	103
CHF	Switzerland	Safe haven	2000Q1–2025Q4	103
JPY	Japan	Safe haven	2000Q1–2025Q4	103
EUR	Eurozone	Control	2000Q1–2025Q4	103
GBP	United Kingdom	Control	2000Q3–2025Q4	101
SEK	Sweden	Control	2000Q1–2025Q4	103

Note. The panel comprises N = 14 countries and 1,434 country-quarter observations.

The oil-exporter group consists of Canada, Mexico, Brazil, and Colombia. These economies share a structural dependence on petroleum as a primary source of foreign-exchange revenues, making them natural points of comparison for Norway in the analysis considering whether the krone's oil sensitivity is distinct from that of comparable oil-exporting economies. Canada is a central case in the oil-currency literature and serves as a natural benchmark for Norway given its status as an advanced open economy with substantial oil-export revenues. Colombia and Mexico derive significant shares of government revenues from petroleum, while Brazil has emerged as a major offshore producer over the sample period. Mexico's oil-export share has declined relative to the other economies in this group, and its inclusion is therefore examined separately in the robustness analysis in Chapter 7.

The control group consists of the Eurozone, the United Kingdom, and Sweden. To identify whether commodity-exposed economies respond differently to oil-price movements, the panel requires a reference group that does not share this exposure. These economies serve that purpose, as primary commodity exports do not constitute a significant share of their foreign-exchange revenues. Sweden is included as a Nordic institutional comparator to Norway, providing a contrast between otherwise similar economies that differ primarily in their degree of petroleum dependence.

Switzerland and Japan form a separate safe-haven group. As documented in Chapter 2, their currencies exhibit well-established safe-haven properties, tending to appreciate against the dollar when global uncertainty rises and risk appetite contracts (Ranaldo & Söderlind, 2010), a pattern associated with structural characteristics such as net foreign asset positions and the size and liquidity of domestic financial markets (Habib & Stracca, 2011). Including them as a distinct group rather than pooling them with the control group ensures that their systematically different response to global risk episodes does not bias the estimates for the reference category.

The other commodity-exporter group consists of Australia, New Zealand, South Africa, and Chile. These economies are significant exporters of metals, minerals, and agricultural products, and their exchange rates have been studied in connection with their own dominant commodity prices in the existing literature (Chen & Rogoff, 2003; Cashin et al., 2004). Here they serve a different role. The empirical motivation draws on evidence that oil-price movements reflect global demand conditions that affect commodity markets broadly (Kilian, 2009), and whether this produces a detectable exchange-rate response among non-oil exporters is an open empirical question the three-group structure is designed to test. Australia is classified in this group rather than the oil-exporter group despite its significant liquefied natural gas exports, as its export base remains broadly diversified across iron ore, coal, gas, gold, and agricultural products.

4.2 Exchange rate data and alignment choices

The analysis uses bilateral nominal exchange rates expressed as units of domestic currency per U.S. dollar, with an increase representing a depreciation of the domestic currency. Bilateral rates against the dollar follow directly from the panel structure, in which the United States serves as the common numeraire, and avoid the aggregation assumptions inherent in trade-weighted effective exchange rates. Nominal rather than real rates are used as the dependent variable because the macroeconomic differentials in the model are designed to capture the fundamental drivers of real exchange rate movements, making deflator adjustments redundant within this framework. All series are sourced from LSEG Datastream as mid-point spot rates.

4.3 Sources, definitions, and construction rules

4.3.1 Interest rates

Short-term interest rate differentials are constructed using three-month market-based rates for each country, expressed as the difference between the domestic rate and the corresponding U.S. rate. Market-based rates are preferred over policy rates because policy rates adjust discretely and may not fully reflect market expectations between central bank meetings. For each country, the most liquid and widely referenced three-month money market benchmark is used, with the three-month horizon chosen to align with the quarterly frequency of the panel and the interest rate differential framework introduced in Chapter 2.

The domestic series vary somewhat in instrument type across countries, reflecting structural differences in national money market conventions and data availability. Most panel countries use unsecured interbank rates, while others use government bill rates or bank-accepted bill rates. These instruments carry marginally different risk characteristics, with unsecured interbank rates bearing a small credit risk premium over sovereign rates, but all are market-determined and continuously priced over a comparable three-month horizon. The differences in risk characteristics are not expected to affect the analysis materially, as they are unsystematic across time and largely absorbed by country and time fixed effects.

Brazil represents a structural exception, as the available series is the SELIC overnight rate, a policy rate set by the central bank rather than a market-determined three-month rate. It is used as a proxy given the absence of a consistent long-run three-month market rate, and this choice is examined in the robustness analysis in Chapter 7. All interest rate series are sourced from LSEG Datastream, with the exception of Colombia's three-month interbank rate which is sourced from FRED. None of the interest rate series are seasonally adjusted.

4.3.2 Consumer prices

Inflation differentials are constructed using national consumer price indices for each country, expressed as the difference between the log change in the domestic price level and the corresponding U.S. rate. Headline CPI is used throughout rather than core or underlying inflation measures, as purchasing power parity is defined in terms of general price levels and not measures that exclude volatile components such as energy and food, and headline indices represent the broadest available measure of consumer price developments in each country. The eurozone is an exception, for which the Harmonised Index of Consumer Prices published by Eurostat is used. As a monetary union that is not a single country, the eurozone has no national CPI, and the HICP is the only available aggregate price measure for the area as a whole. All consumer price series are sourced from LSEG Datastream and are not seasonally adjusted.

4.3.3 Output growth

GDP growth differentials are constructed using GDP in constant prices, seasonally adjusted, expressed as the difference between the log change in domestic output and the corresponding U.S. rate. All series are sourced from LSEG Datastream.

Total GDP is used for all countries in the panel, including Norway. Mainland GDP is the conventional output measure for Norway, but the choice here follows from the identification strategy. Oil prices affect commodity-exporting economies in part through their effect on domestic real activity, since higher oil prices raise petroleum-related output and feed through to aggregate production. Including total GDP as a control allows this activity-based channel to account for the share of exchange-rate variation that runs through real activity, so that any residual oil effect captures something operating beyond it. A natural concern is that Norway shows a stronger oil response because oil constitutes a larger share of GDP than in the other panel economies. Total GDP addresses this specific concern by absorbing the larger activity response that follows from this higher share. A significant Norway-oil interaction therefore reflects oil sensitivity that remains after controlling for any oil effect captured by GDP. As a robustness check, Chapter 7 re-estimates the model using mainland GDP for Norway.

Colombia's GDP series required special treatment, as no single series provides full coverage from 2000Q1. Two series are spliced at Q1 2005, with the log-difference observation at the splice point set to missing. All GDP series are seasonally adjusted and expressed in constant prices.

4.3.4 Oil price and volatility index

The oil price variable is the Brent crude spot price, expressed in U.S. dollars per barrel and sourced from LSEG Datastream. Brent is used rather than West Texas Intermediate as it serves as the primary international benchmark and is the reference price against which Norwegian crude oil is priced.

The volatility regime indicator is based on the CBOE Volatility Index, which measures the implied volatility of S&P 500 index options and is used here as a proxy for global financial market uncertainty. The VIX is sourced from FRED. A binary high-volatility indicator is constructed by comparing each quarterly VIX average against the 75th percentile of the full sample distribution, with quarters above this threshold identified as high-volatility periods. The choice of the 75th percentile as the threshold is examined in the robustness analysis in Chapter 7.

4.4 Frequency harmonization and transformations

The analysis is conducted at quarterly frequency, consistent with the focus on macroeconomic fundamentals. Quarterly data provides a coherent framework for combining financial and macroeconomic variables that are naturally observed at monthly or quarterly intervals and reduces the influence of microstructure noise that dominates high-frequency exchange-rate movements.

Quarterly observations for exchange rates, interest rates, consumer prices, oil prices and GDP are constructed using the mid-month convention, taking the observation at the midpoint of the second month of each quarter, corresponding to February, May, August, and November. This provides a single point-in-time observation that avoids end-of-period distortions and ensures consistent timing across these variables. VIX is constructed as the quarterly average of daily observations, as it serves as a binary regime classifier that captures whether the quarter as a whole reflects elevated stress conditions.

The exchange rate and the oil price enter the model as quarterly log-differences of their levels. Inflation and GDP enter as differentials, each constructed as the difference between the domestic quarterly log-difference (the inflation rate or GDP growth rate) and the corresponding

U.S. rate. Interest rate differentials are expressed as percentage point differences in levels, consistent with their role as linear approximations of expected returns in the UIP framework

4.5 Descriptive statistics and stylized facts

This chapter presents the descriptive statistics of the underlying dataset.

4.5.1 Data analysis

Table 4.3: Descriptive statistics by country group

Variable	**Mean**	**Std. dev.**	**N**	**Corr. with oil**
Norway	0.0020	0.0578	103	-0.415
Oil commodity	0.0058	0.0641	412	-0.454
Other commodity	0.0035	0.0658	406	-0.369
Safe haven	-0.0017	0.0473	206	0.021
Control	0.0001	0.0493	307	-0.310
Full panel	0.0026	0.0590	1434	-0.293
Brent oil (Δlog Brent)	0.0091	0.2203	103	1
VIX, full sample	19.80	7.42	103	-
VIX, low-VIX quarters	16.49	3.38	77	-
VIX, high-VIX quarters	29.61	7.50	26	-

Note. Means and standard deviations are computed across country-quarter observations within each group, measured against the dollar. VIX is reported in levels since it enters the model as a regime indicator.

The Norway row shows an average quarterly change in the krone against the dollar of 0.0020 in log terms, close to zero and absorbed by the country fixed effect, so it carries no information for the regression. What the row contributes is the standard deviation of 0.0578 and the raw correlation with Brent of negative 0.415. The negative correlation shows that the oil-krone relationship is already visible before any controls are applied.

Figure 4.1: Brent oil price against the USD/NOK exchange rate

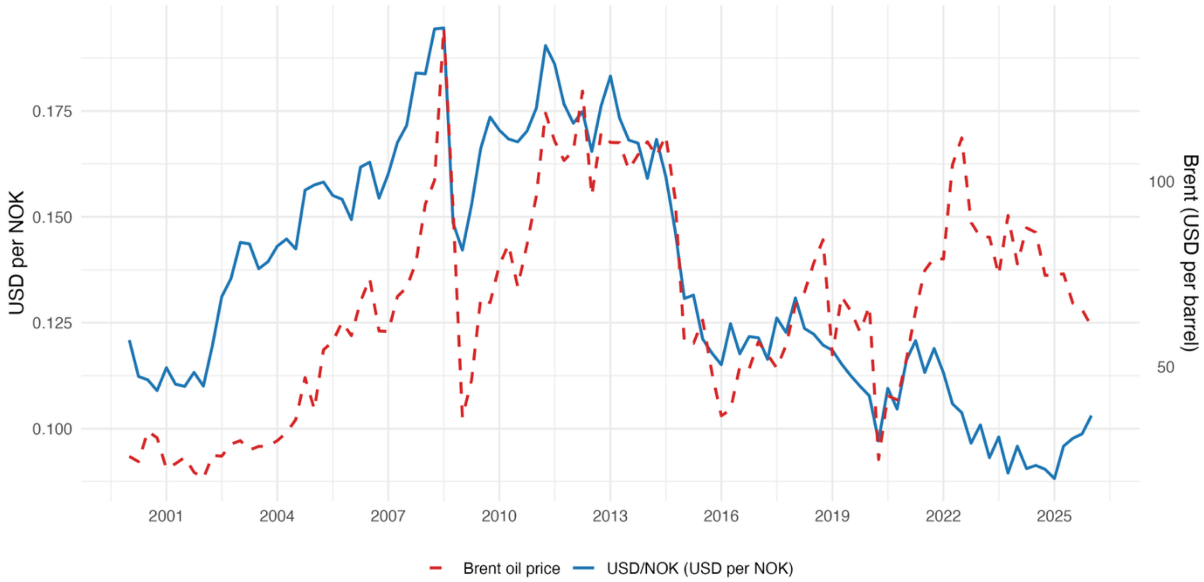


Figure 4.1 plots the Norwegian krone against the Brent crude oil price over the sample period. The two series display a broadly positive co-movement over much of the period, with the krone strengthening as oil prices rose through the mid-2000s and weakening sharply when oil prices collapsed in 2014–2016 and again in 2020. The relationship is not uniform, however. The krone depreciated substantially in the early part of the sample despite stable oil prices, and the post-2022 period shows a persistent weakening of the krone even as oil prices remained elevated.

Figure 4.2: Pearson correlation of groups against oil

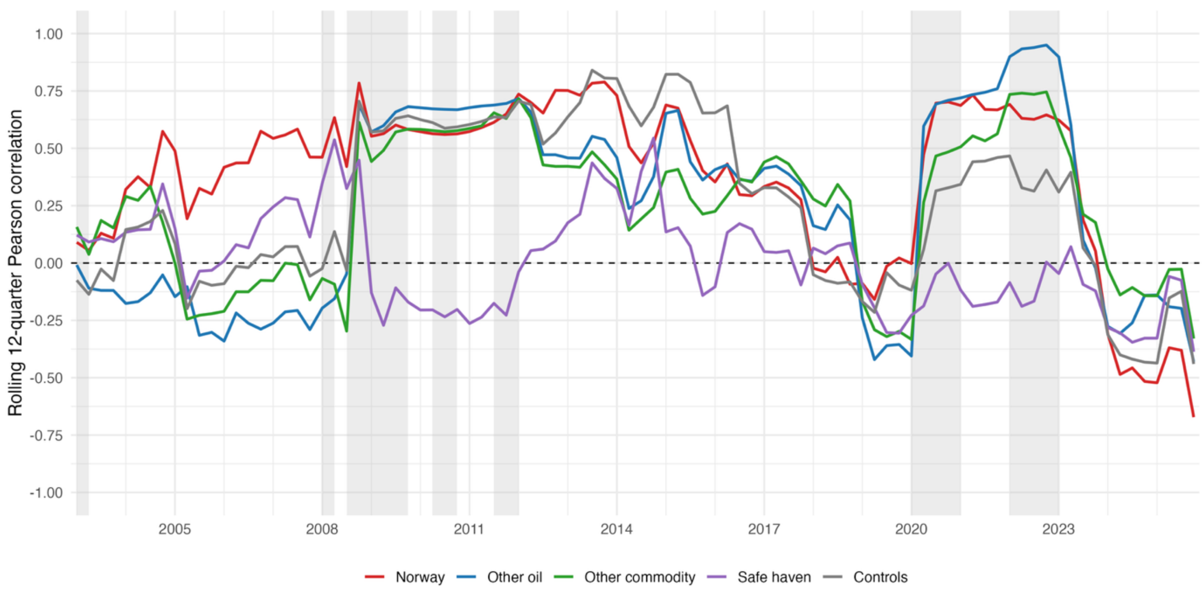


Figure 4.2 examines whether this co-movement is specific to Norway or reflects a broader pattern across currency groups. The figure plots the rolling 12-quarter Pearson correlation between each group's average exchange-rate change and the Brent oil price, with high-volatility

quarters shaded in grey. Several patterns are evident. Norway's correlation with oil is consistently among the highest across groups for most of the sample, though it converges toward other oil-exporting currencies during the financial crisis and the COVID-19 shock. The safe-haven group displays systematically lower and frequently negative correlations with oil across the sample period. All groups exhibit substantial time variation in their oil correlations, with correlations generally rising during high-volatility periods and declining sharply in the most recent part of the sample. These raw correlations reflect both oil-specific dynamics and general movements in the U.S. dollar, as all exchange rates are measured bilaterally against the dollar.

4.5.2 Correlation structure and multicollinearity analysis

Beyond these bivariate patterns, the regressors are also checked for multicollinearity before estimation. Pairwise correlations among the macroeconomic differentials are mostly low, with the largest being between the interest-rate and inflation differentials at 0.47. Among the macroeconomic fundamentals, generalised variance inflation factors are 5.75 (interest-rate differential), 2.43 (inflation differential), and 1.55 (GDP-growth differential), with the highest above the moderate threshold of VIF > 5 but well below the conventional severe threshold of VIF > 10 (Kutner et al., 2005). The oil interaction terms display elevated GVIFs because they share the common Δoil_t component, a necessary feature of testing oil-effect heterogeneity across regimes. Coefficient estimates therefore remain unbiased, with somewhat reduced precision in the interest-rate differential. Full diagnostics are reported in robustness *Table A.1* and *Table A.2.*

5 Econometric Methods and Identification

This chapter explains how the specifications developed in Chapter 3 are estimated, how the identification strategy is justified, and how the research hypotheses are operationalised as formal statistical tests. The structure mirrors the empirical workflow. Section 5.1 sets out the identification strategy and the remaining endogeneity concerns. Section 5.2 discusses the inference procedures used to construct standard errors and confidence intervals. Section 5.3 reports the residual diagnostics that motivate the inference choices. Section 5.4 maps each research hypothesis to a specific statistical test and addresses identification limits at the group-coefficient level.

5.1 Identification and remaining endogeneity concerns

The empirical literature on oil prices and exchange rates has developed along two broadly complementary lines. Structural VAR approaches, such as Kilian and Zhou (2022), identify specific oil supply, oil demand, exchange rate, and interest rate shocks through sign, zero, and narrative restrictions, and trace their effects on macroeconomic variables. Reduced-form commodity-currency studies, such as Chen and Rogoff (2003) on commodity currencies and Bodart et al. (2012) on long-run commodity-price relationships in real exchange rates, instead document systematic relationships between commodity prices and exchange rates across countries without identifying the structural shocks driving commodity prices. This thesis adopts the latter approach.

The identifying variation exploits heterogeneous cross-country exposure to a common factor, following the logic of Lustig et al. (2011) that differences in exposure to global factors are central to understanding currency dynamics.

The inclusion of time fixed effects is central to this design. All movements that are common to the panel in a given quarter are absorbed non-parametrically. These include the average response to changes in the Brent oil price, the U.S. dollar, international risk sentiment, and any global macroeconomic or financial shock that affects all sample currencies symmetrically. The estimated interaction coefficients therefore cannot be driven by the raw co-movement between oil prices and the U.S. dollar, nor by any common global factor that happens to move with oil. They reflect only the differential response of commodity-exposed currencies to that common oil-price movement, measured relative to the omitted reference group.

First concern: bidirectional causality between oil prices and exchange rates. Structural VAR approaches such as Kilian and Zhou (2022) address this directly by decomposing oil-price movements into structural components. The reduced-form approach adopted here cannot make this distinction, and a panel coefficient on oil cannot fully separate the effect of oil prices on exchange rates from reverse channels through dollar denomination. The countries in the sample are price-takers in the global oil market, which limits but does not eliminate this concern. The reduced-form claim being tested is whether commodity-exposed currencies respond differently to the same common oil-price movement, not whether oil causes exchange rate movements in a structural sense.

Second concern: the regime structure and global risk. The VIX-based regime indicator does not control for global risk in the conventional sense. Common volatility conditions affecting all currencies symmetrically are absorbed by the time fixed effects. The φ_g coefficients instead identify whether the comparative oil sensitivity of each group shifts between low- and high-volatility environments, which is precisely the object of interest under Hypothesis 4.

Third concern: omitted variable bias from time-varying country-specific factors. Oil prices may be correlated with variables not included in the model, such as country-specific trade composition, commodity-related fiscal flows, or terms-of-trade movements beyond those captured by the macroeconomic differentials. If these factors also influence exchange rates and differ systematically across commodity-exposed and less-exposed economies, the estimated oil coefficients may partly reflect their influence rather than oil exposure alone. They should therefore be interpreted as conditional correlations.

A final consideration concerns frequency. The use of quarterly data implies that oil-exchange rate relationships operating primarily at higher frequencies may not be captured as sharply by the estimated interactions (Ferraro et al., 2015). The choice is consistent with the focus of the analysis on the slower fundamental drivers of exchange rates. The HighVIX regime indicator follows the same logic. It captures the broader market environment within each quarter, which keeps it conceptually compatible with the quarterly framework.

5.2 Inference

The three panel specifications in equations (3.2)-(3.4) are estimated by ordinary least squares using the fixest package in R (Bergé et al., 2026). Inference must account for the fact that residuals within each country are likely to be serially correlated over time. Exchange-rate movements, interest-rate differentials, inflation dynamics, and macroeconomic conditions all evolve persistently, so unobserved factors that affect a currency in one quarter tend to affect it in subsequent quarters as well. If this within-country dependence is ignored, standard errors are understated because the effective number of independent observations per currency is overstated.

To account for this, standard errors are clustered at the country level, following the empirical framework of Habib and Stracca (2011) and applying the cluster-robust variance estimator discussed in Cameron and Miller (2015). Country clustering allows for arbitrary serial correlation and heteroskedasticity within each country's time series. The two-way fixed-effects structure of equations (3.2)-(3.4) absorbs time-invariant country characteristics and common time-varying factors. For the auxiliary country-by-country specification in equation (3.1), the panel structure is absent, and inference is conducted on a single time series for each currency. Standard errors are computed using the HAC procedure of Newey and West (1987), which adjusts the variance estimator to account for heteroskedasticity and serial correlation in the time-series residuals.

5.3 Residual diagnostics

Beyond the choice of robust inference procedure, residual diagnostics are used to assess whether the residual structure is consistent with the assumptions that motivated that choice. The diagnostics do not guide the choice of method but instead document whether the assumed residual properties hold in the data.

The residual from the preferred specification is the part of quarterly exchange-rate changes that remains unexplained after accounting for macroeconomic differentials, oil-price interactions, country fixed effects, quarter fixed effects, and volatility-regime interactions. Several features of the residual structure could undermine inference, including misspecification of the linear functional form, serial correlation within countries, marked heteroskedasticity, departures from normality, and disproportionate influence from particular observations on the estimates. The diagnostic checks therefore focus on functional form, serial correlation, heteroskedasticity, distributional shape, and influential observations.

Of these checks, one is a formal hypothesis test while the others are based on visual inspection. The Breusch-Godfrey test for panel models (Wooldridge, 2010) is used to test for serial correlation in the residuals. Functional form, heteroskedasticity, distributional shape, and influential observations are assessed through residual plots. Results are reported in Section 6.3, with the corresponding figures in Appendix A.

These diagnostics document the residual structure rather than verify every classical OLS assumption, several of which are unlikely to hold exactly in a macro-financial panel.

5.4 Operationalisation of hypothesis tests

The research hypotheses set out in Section 3.1 are evaluated using coefficient estimates and standard errors from the panel regressions. For hypotheses that concern linear combinations of coefficients, a Wald test is computed using the estimated variance-covariance matrix. Table 5.1 maps each hypothesis to the specific coefficient or linear restriction being tested. In each case, the restriction reflects the null hypothesis, which states the opposite of the research hypothesis, and rejection of the null is therefore interpreted as evidence consistent with the research hypothesis.

Table 5.1: Mapping of research hypotheses to formal test procedures

Hypothesis	Restriction (H_0)	Statistical test	Model
H1	$E[d_i] = 0$	*Mean Group cross-check*	*eq. (3.1)*
	$\theta_N = \theta_O = \theta_C = 0$	*Joint Wald F-test*	*eq. (3.3)*
H2	$\theta_N = 0$	*One-sided t-test*	*eq. (3.3)*
	$\theta_O = 0$	*One-sided t-test*	*eq. (3.3)*
	$\theta_O - \theta_C = 0$	*Wald test on linear combination*	*eq. (3.3)*
	$\theta_N - \theta_C = 0$	*Wald test on linear combination*	*eq. (3.3)*
H3	$\theta_N - \theta_O = 0$	*Wald test on linear combination*	*eq. (3.3)*
H4	$\varphi_N = \varphi_O = \varphi_C = 0$	*Joint Wald F-test*	*eq. (3.4)*

Note. H1 is tested through two complementary procedures (panel and country-by-country). H2 consists of four separate restrictions, evaluated jointly in the verdict. H3 and H4 are single restrictions. F-tests in H1 and H4 are two-sided; t- and Wald-tests in H2 and H3 are one-sided.

The mapping of hypotheses to specifications in Table 5.1 reflects the substantive claim of each hypothesis. H1, H2, and H3 are formulated as unconditional analyses and are therefore tested on Model 2, where the group-level oil coefficients capture average sensitivity over the full sample. H4 concerns how oil sensitivity differs between low and high volatility periods and is therefore tested on Model 3, where the φ_g shifts are identified. The hypothesis tests follow the same logical progression as the model specifications themselves, testing the unconditional and conditional dimensions of oil sensitivity separately. The formal hypothesis tests should be read alongside the full discussion of the regression results in Chapter 6.

The H1 and H4 joint Wald F-tests exclude the safe-haven coefficients θ_S and φ_S. The safe-haven coefficients are reported descriptively as controls that prevent counter-cyclical safe-haven behaviour from contaminating the control reference benchmark, rather than in the basis for formal hypothesis tests. The group-level coefficients for the other oil and non-oil commodity exporters are identified from four countries each, which is small but reportable.

6 Results

This chapter reports the empirical findings of the thesis. First, it analyses the main regression and the included variables to assess how well the models capture the effect of oil on the Norwegian krone. Then, it evaluates the results against the four research hypotheses set out in Chapter 3 using the statistical tests established in prior sections. Lastly, it examines the residuals to assess the robustness of the results.

6.1 Main regression results

Table 6.1 presents the estimates from the three panel specifications. The estimates display three patterns that the subsequent discussion examines in turn. First, the macroeconomic controls are stable in sign and magnitude across the three models, indicating that the oil-related extensions add to the baseline rather than displace it. Second, the explanatory power increases progressively from the macroeconomic baseline through the unconditional oil interactions and into the regime-conditional specification. Third, the group-level oil coefficients differ in ways that bear directly on the krone's position within the panel and on the role of the volatility regime.

Table 6.1: Estimated coefficients for equations (3.2), (3.3), and (3.4)

Variable	**Model 1**	**Model 2**	**Model 3**
Interest-rate differential	−0.0031***	−0.0032***	−0.0032***
	(0.0008)	(0.0008)	(0.0008)
Inflation differential	+0.999***	+0.970***	+0.998***
	(0.251)	(0.259)	(0.259)
GDP-growth differential	−0.164	−0.099	−0.064
	(0.163)	(0.138)	(0.127)
Norway × oil (θ_N)	—	−0.0459***	−0.0479***
		(0.010)	(0.009)
Other oil × oil (θ_O)	—	−0.0393**	+0.0181
		(0.015)	(0.017)
Other commodity × oil (θ_C)	—	−0.0341	+0.0043
		(0.020)	(0.016)
Safe haven × oil (θ_S)	—	+0.0678**	+0.0258*
		(0.025)	(0.014)
Norway × oil × HighVIX (φ_N)	—	—	+0.0063
			(0.029)
Other oil × oil × HighVIX (φ_O)	—	—	−0.0892*
			(0.042)
Other commodity × oil × HighVIX (φ_C)	—	—	−0.0616*
			(0.031)
Safe haven × oil × HighVIX (φ_S)	—	—	+0.0662*
			(0.034)
Country fixed effects	Yes	Yes	Yes
Quarter fixed effects	Yes	Yes	Yes
Group × HighVIX dummies (γ_g)	—	—	Yes
Observations	1,434	1,434	1,434
R^2	0.518	0.537	0.589
Within-R^2	0.032	0.073	0.100
Adjusted within-R^2	0.022	0.063	0.089

*Note: Country-clustered standard errors in parentheses. Significance levels are: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$, two-sided.*

6.1.1 Baseline macro fundamentals model

The macroeconomic controls behave consistently across the three specifications and together provide a baseline against which the oil-related extensions can be evaluated. The interest-rate differential carries a negative coefficient near −0.003, significant at the 1 per cent level in all three models. As an economic reading of the coefficient, a one-percentage-point increase in the domestic short rate relative to the U.S. rate is associated with approximately 0.3 per cent appreciation of the domestic currency against the dollar in the same quarter. The sign runs counter to uncovered interest parity, which would predict a positive coefficient near one, and is consistent with the empirical forward-premium evidence reviewed in Section 2.2.

The inflation differential carries a positive coefficient close to one, also significant at the 1 per cent level. As an economic reading of the coefficient, a one-percentage-point higher domestic inflation rate relative to the United States is associated with approximately a 1 per cent depreciation of the domestic currency against the dollar in the same quarter. Both sign and magnitude align with the prediction of relative purchasing power parity, and the proximity of the coefficient to unity suggests that exchange-rate adjustments approximately offset inflation differentials in the data.

The GDP-growth differential is negative but statistically indistinguishable from zero in any specification. The lack of statistical significance is itself notable, since GDP growth has no detectable explanatory power for exchange-rate movements once interest rates and inflation are accounted for.

6.1.2 Incremental role of oil

With the heterogeneous oil interactions in Model 2, the within-R^2 rises from 0.032 to 0.073, indicating that oil-price exposure contributes explanatory power beyond the macroeconomic fundamentals. The four group-level oil coefficients capture the unconditional oil sensitivity for each group, before the regime decomposition introduced in Model 3 is examined separately in Section 6.1.3. The most negative coefficient is for Norway (θ_N = −0.0459), followed by the four other oil exporters (θ_O = −0.0393) and the four non-oil commodity exporters (θ_C = −0.0341), with the safe-haven group taking the opposite sign (θ_S = +0.0678). The gradient in statistical significance follows the gradient in magnitude, with Norway's coefficient significant at the 1 per cent level, the other-oil and safe-haven coefficients at the 5 per cent level, and the other-commodity coefficient not statistically significant.

Because quarter fixed effects absorb the common oil-price movement, these coefficients should be read relative to the omitted diversified control group. For Norway, a 1 per cent Brent increase implies approximately 0.0459 per cent stronger NOK appreciation against the dollar relative to the control group in the same quarter. Quarterly log changes in Brent have a standard deviation of 0.22 (*Table 4.3*), so a one-standard-deviation Brent shock translates into a relative NOK appreciation of roughly 1.0 per cent.

The safe-haven group moves in the opposite direction, consistent with these currencies responding through a different channel from the commodity-linked groups. Whether this unconditional gradient reflects stable sensitivity across market conditions or averages over different regime-specific patterns is examined in Section 6.1.3.

6.1.3 State dependence and volatility regime results

Model 3 decomposes the unconditional oil sensitivity from Model 2 into a low-volatility coefficient θ_g and a regime-shift coefficient φ_g that captures the change in slope when the $HighVIX$ indicator equals one. The implied high-volatility slope $\theta_g + \varphi_g$ is the sum of the two, and its statistical significance is assessed through a cluster-robust Wald test on the linear restriction $\theta_g + \varphi_g = 0$. *Table 6.2* reorganises the Model 3 estimates from *Table 6.1* by group and reports the implied high-volatility slope with significance markers from the Wald test. The within-R^2 rises from 0.073 in Model 2 to 0.100 in Model 3, indicating that the regime decomposition contributes substantial additional explanatory power beyond the unconditional specification.

For Norway, the low-volatility coefficient θ_N is statistically and economically indistinguishable from the unconditional Model 2 estimate and is significant at the 1 per cent level, while the regime-shift coefficient φ_N is small and not significantly different from zero at any conventional level. The implied high-volatility slope $\theta_N + \varphi_N$ remains close to θ_N and is significant at the 10 per cent level. This indicates that the Norwegian krone is more oil-sensitive than the reference group in both regimes at similar magnitudes and does not display volatility-driven amplification.

For the four other oil exporters and the four non-oil commodity exporters, the picture is reversed. The low-volatility coefficients are close to zero and not significant, while the regime-shift coefficients are negative and significant at the 10 per cent level. The implied high-volatility slopes are negative and significant at the 5 per cent level for the other oil exporters, and at the 10 per cent level for the non-oil commodity exporters. Both groups acquire oil sensitivity only when global volatility is elevated. The unconditional Model 2 coefficients for these two groups therefore pooled a regime in which the response was essentially absent and a regime in which it was clearly negative, a distinction the unconditional specification could not separate.

For the safe-haven group, the pattern is amplified. The low-volatility coefficient (+0.026) is significant at the 10 per cent level with the opposite sign from the commodity groups. The regime-shift coefficient (+0.066) reinforces the response in the same positive direction, producing an implied significant high-volatility slope of +0.092. Safe-haven currencies therefore show a more positive oil sensitivity than the control group, particularly under high-VIX conditions where the implied slope is significant at the 5 per cent level.

The regime decomposition reveals an asymmetric pattern across the four groups. Norway's oil sensitivity is present in both regimes at similar magnitudes, the two non-Norway commodity groups acquire oil sensitivity only in high-volatility periods, while the safe-haven currencies move in the opposite direction with the pattern strengthening under stress. The formal tests on the regime-shift coefficients are reported in *Table 6.2*.

Table 6.2: Implied total oil effects across volatility regimes by group

State Group	**Low VIX** $\boldsymbol{\theta_g}$	**VIX Change** $\boldsymbol{\varphi_g}$	**High VIX** $\boldsymbol{\theta_g + \varphi_g}$
Norway (NOK)	−0.048***	+0.006	−0.042*
Other oil exporters	+0.018	−0.089*	−0.071**
Other commodity	+0.004	−0.062*	−0.057*
Safe haven	+0.026*	+0.066*	+0.092**

Note: Significance markers for $\theta_g + \varphi_g$ are from Wald tests on the restriction $\theta_g + \varphi_g = 0$. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.2 Hypothesis testing

This section presents the formal hypothesis tests defined in *Table 5.1* for each of the four research hypotheses defined in chapter 3. The tests draw on the regression results from Section 6.1 but go beyond the descriptive analysis by comparing the commodity-exposed groups against each other rather than only against the diversified reference. H1 and H4 take up the broader claims about whether oil matters at all and how its effect varies across volatility regimes, while H2 and H3 examine within-commodity contrasts that are not directly visible in *Table 6.1.*

Table 6.3: Formal hypothesis-test verdicts

Hypothesis	Equation	Restriction (H_0)	Estimate	p-value	Verdict
H1	Eq. 3.3	$\theta_N = \theta_O = \theta_C = 0$	F(3,13) = 8.11	0.003	Reject H_0 at 1%
	Eq. 3.1	$E[d_i] = 0$	−0.058	< 0.001	Reject H_0 at 1%
H2	Eq. 3.3	$\theta_N = 0$	−0.046	< 0.001	Reject H_0 at 1%
	Eq. 3.3	$\theta_O = 0$	−0.039	0.012	Reject H_0 at 5%
	Eq. 3.3	$\theta_O - \theta_C = 0$	−0.005	0.412	Do not reject H_0
	Eq. 3.3	$\theta_N - \theta_C = 0$	−0.012	0.245	Do not reject H_0
H3	Eq. 3.3	$\theta_N - \theta_O = 0$	−0.007	0.334	Do not reject H_0
H4	Eq. 3.4	$\varphi_N = \varphi_O = \varphi_C = 0$	F(3,13) =11.32	< 0.001	Reject H_0 at 1%

Note. Panel p-values are computed with country-clustered standard errors. The joint tests for H1 and H4 are Wald F-tests (two-sided); the remaining rows use t-tests, one-sided for H2 and H3.

6.2.1 Hypothesis 1: Incremental explanatory power of oil

Hypothesis 1 states that oil-price movements add explanatory power for exchange-rate movements beyond the standard macroeconomic fundamentals. The hypothesis is operationalised in two complementary ways. The panel test asks whether the three group-level oil interactions in Model 2 are jointly zero (the null hypothesis), with rejection providing evidence consistent with H1. The country-by-country test asks whether the average oil coefficient across single-country time-series regressions in equation (3.1) differs from zero. The two procedures answer the same research question through different identification strategies, and convergence between them provides consistent evidence on H1.

The panel test evaluates the joint Wald restriction $\theta_N = \theta_O = \theta_C = 0$ in Model 2. The restriction is rejected at the 1 per cent level, with F(3, 13) = 8.11 and p-value 0.003. The Mean Group aggregate of the coefficients from equation (3.1) is −0.058, with p-value below 0.001. Table 6.4 reports the country-level estimates.

Across the nine oil and commodity-exposed currencies, all point estimates are negative, with five significant at the 1 per cent level, one further at the 5 per cent level, and one further at the 10 per cent level. Unlike the panel coefficients in Table 6.1, which are identified relative to the omitted control group, the country-by-country coefficients in equation (3.1) are absolute responses against the dollar, since the single-country specification has no quarter fixed effects to absorb the common oil-price movement. The country-level estimates are descriptive and complementary to the panel evidence, though they cannot separate currency-specific oil sensitivity from common dollar movements that the panel time fixed effects absorb.

Table 6.4: Auxiliary country-by-country oil coefficients

Currency	Group	Oil coef.	HAC SE	t-stat	p-value
NOK	Norway	−0.083	0.023	−3.6	< 0.001
BRL	Other oil	−0.115	0.035	−3.3	< 0.001
CAD	Other oil	−0.078	0.016	−4.85	< 0.001
COP	Other oil	−0.053	0.027	−1.97	0.049
MXN	Other oil	−0.039	0.034	−1.14	0.255
ZAR	Other commodity	−0.109	0.036	−3.06	0.002
AUD	Other commodity	−0.086	0.023	−3.68	< 0.001
NZD	Other commodity	−0.070	0.036	−1.93	0.053
CLP	Other commodity	−0.029	0.018	−1.61	0.107
CHF	Safe haven	−0.016	0.017	−0.95	0.340
JPY	Safe haven	+0.033	0.020	+1.60	0.109
GBP	Diversified	−0.075	0.028	−2.65	0.008
SEK	Diversified	−0.056	0.026	−2.13	0.033
EUR	Diversified	−0.033	0.025	−1.35	0.178

Note. The table reports estimates of the oil-price coefficient from country-specific time-series regressions corresponding to equation (3.1). Standard errors are Newey–West (HAC).

The verdict is that both procedures reject the zero-oil null. The convergence between the panel and country-by-country evidence is consistent with H1.

6.2.2 Hypothesis 2: Heterogeneity in oil sensitivity across currency groups

Hypothesis 2 states that oil commodity currencies exhibit greater oil sensitivity than other commodity and control currencies. The hypothesis is operationalised through four restrictions on the Model 2 oil coefficients reported in Table 6.1. Two restrictions test whether Norway is more oil-sensitive than the diversified reference and the non-oil commodity group, and two further restrictions test the same comparisons for the pooled group of other oil exporters. The panel specifications treat Norway and the other oil exporters as separate groups, consistent across all hypotheses, so the class-level claim of H2 is tested through restrictions on each class member rather than through a pooled class-level coefficient.

As already shown in the Model 2 regression in Table 6.1, both oil commodity groups reject the zero-oil null relative to the diversified reference. The within-commodity contrasts, however, are new linear combinations not directly reported in the regression. The estimate for Norway is $\theta_N - \theta_C = -0.012$ with one-sided p-value 0.245, and for the other oil exporters $\theta_O - \theta_C = -0.005$ with one-sided p-value 0.412. Neither contrast rejects, so neither oil commodity group is statistically distinguishable from the non-oil commodity exporters in the unconditional specification.

The verdict is that two of the four restrictions reject the zero-oil null. Both oil commodity groups are distinct from the diversified reference, but neither is distinguishable from the non-oil commodity exporters in the unconditional specification. The evidence is consistent with the part of H2 that contrasts oil commodity currencies with the diversified reference, but not with the part that contrasts them with the non-oil commodity group.

6.2.3 Hypothesis 3: Norway's distinct oil sensitivity

Hypothesis 3 states that the oil sensitivity of the Norwegian krone exceeds that of other oil-exporting currencies. This is a direct test of one of the thesis's central questions, namely whether Norway is genuinely distinct within the oil commodity class, or whether it is best understood as one oil exporter among others. While H1 and H2 establish that oil matters and that oil commodity currencies differ from the diversified reference, neither result contains a direct comparison between Norway and the other oil exporters. The Norway and Other oil coefficients in Table 6.1 are each estimated against the diversified reference, so the contrast between them is a new linear combination not directly visible in the regression output. The hypothesis is operationalised as the restriction $\theta_N - \theta_O = 0$ on the Model 2 oil coefficients.

The estimate is $\theta_N - \theta_O = -0.007$ with one-sided p-value 0.334. The restriction does not reject the zero-difference null, indicating that Norway and the other oil exporters are not statistically distinguishable when oil sensitivity is averaged over the full sample.

The verdict is that the unconditional specification does not provide evidence consistent with H3. Averaged across the full sample, the Norwegian krone is not more oil-sensitive than the pooled group of other oil exporters. The unconditional test averages oil sensitivity across volatility regimes, however, and Section 6.1.3 documented that the four currency groups respond very differently to the volatility environment. The question of whether Norway is distinct therefore re-emerges in regime-decomposed form under H4.

6.2.4 Hypothesis 4: State dependence in oil sensitivity

Hypothesis 4 states that the exchange-rate effect of oil prices varies across volatility regimes. The hypothesis is operationalised as a joint Wald F-test on the three regime-shift coefficients in Model 3 reported in Table 6.2, asking whether φ_N, φ_O, and φ_C are jointly zero. A rejection of the null indicates that oil sensitivity differs systematically between low-volatility and high-volatility quarters in at least one commodity-exposed group.

The joint test rejects the null at the 1 per cent level, with $F(3, 13) = 11.32$ and p-value below 0.001. The regime-shift coefficients reported in Table 6.2 show that the rejection is driven by the other oil and non-oil commodity exporters, while the Norwegian regime-shift coefficient is small and statistically indistinguishable from zero.

The verdict is that the joint zero-shift null is rejected. The evidence is consistent with H4 as an aggregate claim, while the regime-shift coefficients in Table 6.2 show heterogeneity within the commodity-exposed groups.

Taken together, the regression analysis in Section 6.1 and the formal hypothesis tests in H1 to H4 indicate that the commodity-exposed groups respond differently to oil prices across the volatility regimes. In low-volatility quarters, the Norwegian krone is the only commodity-exposed currency whose oil coefficient is statistically distinct from the diversified reference, while the other oil and non-oil commodity exporters become substantially more oil-sensitive only when the VIX is elevated. The unconditional hypothesis tests show no statistically distinguishable differences within the commodity class when sensitivity is averaged across regimes. This pattern suggests that Norway's distinct profile may lie in the structure of its oil response rather than in its average magnitude. This pattern motivates a closer analysis of how Norway's oil sensitivity compares with the other commodity-exposed groups within each volatility regime, taken up in the next section.

6.2.5 The Norway contrast across volatility regimes

This section tests how Norway compares with the other commodity-exposed groups within each volatility regime. The four Wald restrictions on the Model 3 coefficients in Table 6.2 contrast Norway with the other oil exporters and with the non-oil commodity exporters in low-volatility and high-volatility quarters.

Table 6.5: H3 regime-conditional contrasts

Regime	**Equation**	**Restriction**	**Estimate**	***p*-value**
Low VIX	Eq. 3.4	$\theta_N - \theta_O = 0$	−0.0660	< 0.001
Low VIX	Eq. 3.4	$\theta_N - \theta_C = 0$	−0.0522	< 0.001
High VIX	Eq. 3.4	$(\theta_N + \varphi_N) - (\theta_O + \varphi_O) = 0$	+0.0296	0.226
High VIX	Eq. 3.4	$(\theta_N + \varphi_N) - (\theta_C + \varphi_C) = 0$	+0.0158	0.448

Note. Low VIX tests are one-sided, High VIX tests are two-sided. Cluster robust standard errors.

The results show a clear regime-conditional pattern. In low-volatility quarters, the Norwegian krone is statistically distinct from both the other oil exporters ($\theta_N - \theta_O$ = −0.066, p < 0.001) and the non-oil commodity exporters ($\theta_N - \theta_C$ = −0.052, p < 0.001), with both contrasts rejecting at the 1 per cent level. In high-volatility quarters, neither contrast is significant, with implied total differences of $(\theta_N + \varphi_N) - (\theta_O + \varphi_O)$ = +0.03 (p = 0.226) and $(\theta_N + \varphi_N) - (\theta_C + \varphi_C)$ = +0.016 (p = 0.448). The Norway-versus-others gap thus effectively disappears under elevated volatility. The verdict is therefore that the Norwegian krone is significantly distinct from the broader commodity class only in tranquil markets.

6.2.6 Overall Research hypothesis verdicts.

The four hypothesis tests provide a structured account of how oil prices relate to the commodity-exposed currencies in the panel. H1 establishes that oil-price movements add explanatory power beyond the macroeconomic baseline, with both the panel joint test and the country-by-country Mean Group procedure rejecting the zero-oil null. H2 finds partial support. Both Norway and the other oil exporters are statistically distinct from the diversified reference, but neither is distinguishable from the non-oil commodity exporters in the unconditional specification. H3 does not find evidence consistent with Norway being distinct from the other oil exporters when oil sensitivity is averaged over the full sample, though Section 6.2.5 shows that this masks a regime-dependent pattern. H4 rejects the joint zero-shift null, with the regime structure driven by the other oil and non-oil commodity exporters rather than by Norway.

6.3 Residual diagnostics

The hypothesis tests in Section 6.2 rely on a robust covariance procedure. The residual diagnostics in this section confirm that the procedure choice is appropriate and identify any features of the residual structure that would qualify the substantive conclusions. The diagnostic categories considered are functional form, heteroskedasticity, serial correlation, residual distribution, and influential observations, with the related figures reported in Appendix A.

6.3.1 Functional form and heteroskedasticity

The residuals-versus-fitted plot in *Figure A.3* does not display a clear funnel shape or systematic increase in variance across fitted values, indicating that heteroskedasticity is not a meaningful problem in the data. The trend line is flat across the range of fitted values, with only a slight upward drift in the upper tail, where observations are few. This indicates that the linear specification captures the systematic relationship between regressors and the dependent variable without evidence of curvature. The residual boxplots in *Figure A.2* show some variation in residual dispersion across countries, but no currency exhibits dramatically wider spread than the rest. Such variation is expected in a panel of bilateral exchange rates and is accommodated by the country-clustered standard errors used throughout the chapter.

6.3.2 Serial correlation

Classical OLS inference requires that the error terms are serially uncorrelated. In a panel of exchange rates and macroeconomic fundamentals, where the included regressors follow slow-moving patterns over time, any unobserved factors omitted from the model are likely to share the same persistence and induce dependence between successive residuals within each country. If such autocorrelation is present and unaccounted for, classical standard errors are downward-biased, and the resulting t-statistics over-reject the null hypothesis. The Breusch-Godfrey-Wooldridge panel LM test returns a statistic of 119.1 with p-value 0.063, borderline at the 5 per cent level and consistent with the presence of some autocorrelation. To account for this, the country-clustered standard errors used throughout this chapter are robust to arbitrary serial correlation.

6.3.3 Residual distribution

Strong departures from normality in the residuals can signal model misspecification or extreme observations that require closer inspection. The histogram in *Figure A.4* and the Q-Q plot in *Figure A.5* show that the residuals are symmetric around zero with a more concentrated peak and heavier tails than the normal distribution. The deviations in both tails reflect single-quarter exchange rate movements during periods of financial stress, discussed further in Section 6.3.4. Cluster-robust inference does not rely on normality, so the heavy tails do not affect the validity of the inference reported in this chapter.

6.3.4 Outliers and influential observations

The residual time series in *Figure A.1* identifies three clusters of large residuals, a spike in 2002-2003, a spike during the global financial crisis in 2008-2009, and a spike at the COVID-19 onset in 2020 Q1. The boxplot in *Figure A.2* shows that extreme outliers occur across all currencies in the panel, with some variation in dispersion between countries. The Norwegian krone residuals are well-centred with a narrow interquartile range and few extreme observations, so the country whose oil sensitivity is the focus of the thesis does not exert disproportionate leverage on the estimates.

The group-specific oil-by-volatility interactions in Model 3 are designed to capture how the sensitivity of exchange rates to oil prices depends on the global volatility regime, with the coefficients estimated from the contrast between low-VIX and high-VIX quarters. The residuals nevertheless show elevated values in crisis periods, indicating that the regime structure absorbs part of the crisis-period variation but does not fully account for it. These large residuals are not statistical noise or data errors but identifiable economic events that reflect dynamics outside the macroeconomic fundamentals the model is designed to capture.

Taken together, the diagnostics confirm that the panel is well-specified for the research questions of the chapter. The one feature that motivates further investigation is the concentration of large residuals in crisis periods. Chapter 7 probes this through subsample re-estimation that excludes the GFC, the 2014 oil-price collapse, and the COVID-19 episodes in turn.

7 Robustness

This chapter examines whether the central findings of Chapter 6 are robust to alternative modelling choices. Those findings are Norway's significant oil sensitivity, the regime-conditional pattern in which other commodity-exposed groups acquire oil sensitivity primarily under elevated volatility, and Norway's distinctness within the commodity class in low-volatility quarters. A finding is judged robust when its sign, statistical significance, and approximate magnitude are preserved across reasonable alternative specifications. Substantial movement in any of these indicates sensitivity to the specification choice. To keep the chapter readable, each variant reports only the coefficients and contrasts relevant to the finding being tested, often Norway's alone, with the full regression output collected in Appendix B.

7.1 Monthly frequency robustness

While the main model targets quarterly relationships, a monthly specification is estimated as a robustness check to verify that the central findings are not specific to quarterly aggregation. Due to data availability, the monthly panel has three limitations. Quarterly GDP growth is replaced by monthly industrial-production growth as the activity proxy. AUD, NZD, and CHF are excluded because their CPI or industrial-production series lack genuine monthly observations. The sample is also shorter, spanning from February 2000 to November 2023. The dataset contains 3,119 country-month observations across eleven currencies.

Table 7.1: Estimated coefficients for equations (3.2), (3.3), and (3.4) at monthly frequency

Variable	**Model 1**	**Model 2**	**Model 3**
Interest-rate differential	−0.0012***	−0.0012***	−0.0013***
	(0.0003)	(0.0003)	(0.0003)
Inflation differential	+0.4735***	+0.4587***	+0.4614***
	(0.1242)	(0.1376)	(0.1371)
Industrial production differential	−0.0293**	−0.0267***	−0.0244***
	(0.0094)	(0.0081)	(0.0075)
Norway × oil (θ_N)		−0.0526***	−0.0560***
		(0.0068)	(0.0041)
Other oil × oil (θ_O)		−0.0361***	−0.0198*
		(0.0105)	(0.0101)
Other commodity × oil (θ_C)		−0.0354***	−0.0187***
		(0.0085)	(0.0057)
Safe haven × oil (θ_S)		+0.0558***	+0.0260***
		(0.0069)	(0.0040)
Norway × oil × HighVIX (φ_N)			+0.0073
			(0.0111)
Other oil × oil × HighVIX (φ_O)			−0.0284
			(0.0201)
Other commodity × oil × HighVIX (φ_C)			−0.0314**
			(0.0115)
Safe haven × oil × HighVIX (φ_S)			+0.0542***
			(0.0110)
Country fixed effects	Yes	Yes	Yes
Monthly fixed effects	Yes	Yes	Yes
Group × HighVIX dummies (γ_g)	—	—	Yes
Observations	3,119	3,119	3,119
R^2	0.490	0.500	0.503
Within R^2	0.013	0.032	0.039
Adjusted within-R^2	0.012	0.030	0.033

Note. Country-clustered standard errors in parentheses. Industrial production growth replaces GDP growth as activity proxy. Significance levels are: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$, two-sided.

The macroeconomic controls behave broadly consistently with the quarterly baseline. The interest-rate and inflation differentials retain their direction and 1 per cent significance. The activity differential, measured as industrial production growth is significant at the 1 per cent level for two of the three models, in contrast to the quarterly specification where the GDP-growth differential was not statistically significant. While the individual coefficients are more precisely estimated at monthly frequency, the within R^2 falls from 0.100 in the quarterly baseline to 0.039 in Model 3, reflecting that monthly exchange-rate variation contains more high-frequency noise that the model's fundamental drivers do not capture.

Table 7.2: Implied total oil effects across volatility regimes, monthly panel

State group	Low VIX $\boldsymbol{\theta_g}$	VIX change $\boldsymbol{\varphi_g}$	High VIX $\boldsymbol{\theta_g + \varphi_g}$
Norway (NOK)	−0.056***	+0.0073	−0.049***
Other oil exporters	−0.020*	−0.028	−0.048**
Other commodity	−0.019***	−0.031**	−0.050***
Safe haven	+0.026***	+0.054***	+0.080***

Note. Significance markers for $\theta_g + \varphi_g$ are from Wald tests on the restriction $\theta_g + \varphi_g = 0$. Significance levels: *** p < 0.01, ** p < 0.05, * p < 0.10.

The monthly panel preserves the magnitude pattern from Chapter 6. Norway has the strongest negative low-VIX coefficient at −0.056 and the only response that is roughly stable across regimes. The other commodity-exposed groups converge toward Norway's magnitude under elevated volatility, with other commodity moving from −0.019 to −0.050 and other oil from −0.020 to −0.048, while the safe-haven group continues in the opposite direction with positive coefficients that strengthen under stress. What differs from the quarterly baseline is the low-VIX coefficient for the non-Norway commodity groups. The monthly specification captures these effects more clearly, and they emerge as significantly negative while the regime amplification on top of this baseline is preserved. Norway's regime shift is again small and not significant, consistent with the stable response found in the quarterly.

7.1.1 H4 retested at monthly frequency

The monthly results in Table 7.2 suggest that the volatility regime effects appear to be captured more sharply. This pattern suggests that the monthly specification may be better suited to identifying state dependence in oil sensitivity, even if it captures less of the variation in exchange rates overall. Since Hypothesis 4 is specifically about how oil sensitivity varies across volatility regimes, this section re-estimates the joint test of H4 at monthly frequency, where the underlying regime structure is more sharply identified.

Table 7.3: Re-estimation of hypothesis 4 at monthly frequency

Hypothesis	Equation	Restriction (H_0)	Estimate	p-value	Verdict
H4	Eq. 3.4	$\varphi_N = \varphi_O = \varphi_C = 0$	F = 77.684	< 0.001	Rejected at 1%

Note. Country-clustered standard errors. The corresponding quarterly result is reported in Table 6.3.

The joint Wald F-test at monthly frequency returns F = 77.68 with p-value below 0.001, rejecting the null that oil sensitivity does not vary across volatility regimes. This supports the research hypothesis that the exchange-rate effect of oil prices varies across volatility regimes. Although the regime-shift coefficients are individually more sharply estimated at monthly frequency, the joint conclusion for H4 is the same across both frequencies.

Lastly, the same regime-conditional tests between groups, as in Section 6.2.5, are estimated with the monthly data, giving the same results, where Norway is statistically different from both other groups in low-VIX regimes, but not distinguishable in high-volatility regimes, as shown in *Table B.11*.

7.2 Alternative volatility-regime definitions

As previously established, the baseline model defines high-volatility quarters using the 75th percentile of the VIX as the regime cut-off. The literature, however, applies different thresholds across studies, reflecting the absence of a single conventional standard. The findings could therefore be sensitive to this specific choice. To assess this, the high-volatility indicator is re-defined at alternative VIX percentiles and absolute levels.

Table 7.4: VIX threshold sensitivity, Norway estimates

Threshold	Low VIX θ_N	VIX change φ_N	High VIX $\theta_N + \varphi_N$	Within R^2
VIX ≥ p50	−0.0441***	−0.002	−0.046**	0.092
VIX ≥ p75 (baseline)	−0.0479***	+0.006	−0.042*	0.100
VIX ≥ p90	−0.0457***	−0.003	−0.049**	0.114

*Note. Additional VIX percentile and absolute-level cut-offs (p80, VIX ≥ 20, 25, 30) gave coefficients in the same range and sensitivity. Country-clustered standard errors. Significance levels: *** p < 0.01, ** p < 0.05, * p < 0.10.**

Norway's coefficients are nearly identical across the three thresholds, despite substantial differences in regime definition. At p50 the high-VIX regime includes the upper half of all quarters, while p90 restricts it to the upper decile. The stability across such different definitions indicates that Norway's oil response is not concentrated at the most extreme volatility quarters. The low-VIX and high-VIX coefficients are also similar within each specification, with insignificant regime-shifts throughout, consistent with the main model's finding that Norway's regime-shift coefficient is essentially zero. This suggests that the absence of regime asymmetry for Norway reflects a stable underlying response rather than the chosen threshold.

7.3 Single country exclusions

Two countries in the panel were flagged in Chapter 4 as potentially problematic. Mexico's declining oil-export share raises questions about its classification as an oil exporter, while Brazil's interest rate is proxied by the SELIC policy rate rather than a market-determined three-month rate. Each is dropped from the panel in turn to test whether the central finding is driven by these countries.

Table 7.5: Single-country exclusions

Specification	**Low VIX** θ_N	**VIX change** φ_N	**High VIX** $\theta_N + \varphi_N$	**N**	**Within** R^2
Mexico excluded	−0.0481***	+0.011	−0.037*	1,331	0.093
Brazil excluded	−0.0488***	+0.006	−0.043*	1,331	0.093

Note. M3 re-estimated with the indicated country dropped. Country-clustered standard errors. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Both single-country exclusions leave Norway's low-VIX coefficient essentially unchanged at the baseline level, with regime-shift coefficients close to zero and high-VIX coefficients within the same band as the baseline. The central finding from Chapter 6 is therefore not driven by Mexico's weaker oil-export classification or by Brazil's SELIC-based interest-rate proxy.

7.4 Crisis-period sensitivity

The baseline panel covers 2000Q1 through 2025Q4, a span that includes several episodes of unusually large oil-price and financial-market movement. The Global Financial Crisis of 2008-2009, the dramatic oil-price collapse beginning in mid-2014 (Baumeister & Kilian, 2016), and the COVID-19 shock of 2020 all introduce large fluctuations into both the dependent variable and the regressors. A natural concern is whether the regime-conditional oil sensitivities documented in Chapter 6 reflect a stable underlying relationship or are instead driven disproportionately by a small number of crisis observations in which oil prices, exchange rates, and the VIX move sharply together. This section addresses that concern by re-estimating the baseline specification with crisis episodes excluded from the sample one at a time. Three exclusions are considered: the Global Financial Crisis, the COVID-19 shock, and the oil-price collapse that began in mid-2014.

Table 7.6: Crisis-period exclusions

Specification	**Low VIX** θ_N	**VIX change** φ_N	**High VIX** $\theta_N + \varphi_N$	**N**	**Within** R^2
Exclude GFC	−0.0479***	−0.019	−0.067***	1,379	0.085
Exclude Oil-Crash	−0.0599***	+0.0191	−0.0408*	1,336	0.107
Exclude COVID	−0.0475***	+0.042	−0.006	1,378	0.085

Note. The GFC window is 2008Q3–2009Q2, the oil-price collapse 2014Q4–2016Q1, and the COVID-19 window 2020Q1–2020Q4. Country-clustered standard errors. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The low-VIX coefficient is stable across all three exclusions, indicating that the central finding is not driven by crisis periods. The regime-shift is also small and statistically insignificant in all three cases, consistent with the absence of regime asymmetry documented throughout the chapter. The high-VIX coefficient, however, varies more, weakening to zero when COVID is excluded, strengthening to 1 per cent significance when the GFC is excluded, and remaining significantly negative but smaller in magnitude when the 2014 oil-price collapse is excluded. This variation reflects the construction of the high-VIX coefficient as the sum of the low-VIX response and the regime-shift. The low-VIX response is stable across exclusions, while the regime-shift takes different signs across the three specifications even though it is insignificant in each. The implied high-VIX sum varies with sample composition, consistent with the broader pattern that the high-VIX channel depends on which stress episodes are included.

7.5 Activity measure for Norway

The baseline uses total Norwegian GDP as the activity measure for Norway, motivated in Section 4.3.3 by the identification strategy of letting the activity channel absorb oil-related variation in real output, so that the residual oil coefficient captures effects beyond the activity channel. Mainland GDP, which excludes the petroleum sector, is the more conventional measure for Norway in domestic analyses. To test whether the choice affects the central finding, the model is re-estimated with mainland GDP replacing total GDP for Norway.

Replacing total GDP with mainland GDP leaves Norway's oil response virtually unchanged. The low-VIX coefficient moves marginally from -0.0479 to -0.0477, and the regime-shift and high-VIX coefficients remain essentially unchanged with the same significance levels. The GDP coefficient itself is insignificant under both measures. The result therefore does not depend on the choice of activity measure for Norway.

7.6 Alternative measures of expectations and uncertainty

The baseline framework uses the VIX as the regime indicator. As outlined in Chapter 2, exchange rates may respond to shifts in expectations, risk appetite, or market narratives. The VIX is well suited to capture this dimension because it is derived from option prices and reflects market participants' forward-looking assessments of near-term risk.

A natural question is whether the VIX adequately captures the forward-looking narrative dimension, or whether an alternative measure constructed through a different method would produce different results. The World Uncertainty Index (WUI; Ahir et al., 2022) is a text-based measure constructed from Economist Intelligence Unit country reports, counting references to uncertainty in analysts' discussions of country outlooks. The WUI captures uncertainty-related narratives through a non-market-based method and is commonly used as a sentiment indicator. If the VIX adequately captures the forward-looking dimension, the WUI-based regime should produce broadly similar results. If not, it could indicate that the VIX fails to capture an important part of the narrative aspect of expectations. To assess this, the Norway regression is re-estimated with the WUI replacing the VIX in the regime classification.

Table 7.7: Alternative regime variables for Norway

Indicator	**θ_N (low WUI)**	**φ_N**	**$\theta_N + \varphi_N$ (high WUI)**
WUI 75 pct.	−0.028***	−0.042***	−0.070***

*Note. The high-uncertainty indicator is constructed at the 75th percentile of the quarterly WUI distribution, matching the baseline threshold for the VIX. Country-clustered standard errors. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.*

The Norway low-stress oil coefficient remains significantly negative at the 1 per cent level under the WUI regime, somewhat smaller in magnitude than the VIX baseline but pointing in the same direction. The high-stress coefficient is also significantly negative under both regimes, and similarly so in the monthly specification from Section 7.1, confirming that the qualitative direction of the relationship is consistent across the three measures.

The main difference lies in the regime-shift, which is essentially zero under the VIX baseline but significantly negative under WUI. The WUI regime may therefore detect a regime asymmetry that the VIX does not, suggesting that the two measures overlap in capturing the overall oil response but differ in how they classify high-stress periods.

7.7 Oil specificity - The Commodity Terms of Trade test

A central question for the interpretation of the baseline results is whether Norway's distinctness reflects exposure to oil specifically or to a broader pattern in which commodity-exporting currencies respond to their own commodity prices. To address this, Brent is replaced by a country-specific commodity terms-of-trade index, which weights commodity-price changes by each country's net trade balance in each commodity. The CTOT index is taken from the IMF Commodity Terms of Trade Database (Gruss & Kebhaj, 2019), which provides country-specific commodity terms-of-trade indices for 182 economies based on country-level trade weights and global commodity prices. Norway's CTOT is dominated by oil and gas, so Norway's response under CTOT should approximate its Brent response. The more interesting test is whether the other commodity-exposed groups respond differently when commodity exposure is measured on a comparable basis across countries rather than against the common Brent benchmark.

The index is constructed for each country i and quarter t as a trade-weighted product of relative commodity prices:

$$CTOT_{i,t} = \prod_{c \in C_i} \left(\frac{P_{c,t}}{P_{c,base}} \right)^{w_{i,c}}$$

where $P_{c,t}$ is the global price of commodity c, $W_{i,c}$ is country i's net export weight in commodity c, and C_i is the set of commodities for which country i trades.

Table 7.8: CTOT specification

Group	**Low-VIX** $\boldsymbol{\theta_g}$	**VIX change** $\boldsymbol{\varphi_g}$	**High-VIX** $\boldsymbol{\theta_g + \varphi_g}$
Norway	−0.656***	+0.345*	−0.311
Other oil exporters	−1.716***	+1.169	−0.547
Other commodity	−1.008***	+0.874	−0.133

*Note. Brent replaced by a country-specific commodity terms-of-trade index. Safe haven is absorbed into the baseline due to limited commodity activity. Country-clustered standard errors. Significance levels: *** p < 0.01, ** p < 0.05, * p < 0.10.*

Norway is significantly more negative than the reference in low-VIX quarters under CTOT, in line with the Brent baseline, but not statistically distinguishable from the reference in high-VIX quarters. The substantive change concerns the other two commodity-exposed groups. Under Brent in the quarterly panel, both other oil exporters and other commodity exporters were not significantly different from the reference in low-VIX quarters and acquired significant negative responses only in high-VIX quarters. Under CTOT, this regime pattern reverses. Both groups are significantly more negative than the reference in low-VIX quarters and are not statistically distinguishable from the reference in high-VIX quarters.

The inversion of the regime pattern for the non-Norway groups suggests that CTOT may capture commodity exposure more fundamentally than Brent for these groups and raises the question of why their oil response would converge toward Norway's magnitude only under elevated volatility.

7.8 Summary of the robustness evidence

The robustness checks in this chapter confirm the central findings of Chapter 6 along the dimensions that matter most for the thesis. Norway's structurally negative oil response in calm market conditions is the most robustly identified feature of the relationship, holding across alternative frequencies, sample periods, volatility thresholds, country compositions, crisis episodes, activity measures, and alternative measures of forward-looking uncertainty. When oil exposure is replaced by a country-specific commodity terms-of-trade index, all commodity-exposed groups display a significant low-stress response to their own commodity baskets. Norway's distinctness under the Brent specification may therefore reflect its specific exposure to oil prices rather than a uniquely strong commodity response.

The high-VIX coefficient is broadly stable in magnitude across robustness specifications, clustering around the baseline value. Its statistical significance varies somewhat, with some deviation toward zero observed when certain periods are excluded. The regime-shift point estimate varies in sign across specifications but is rarely individually significant.

Together, the results support a reading in which Norway's oil sensitivity is a robust structural feature of the krone, of comparable magnitude across both high and low stress regimes.

8 Discussion

The following chapter takes a step back from the empirical documentation in Chapters 6 and 7 to consider what the findings may reflect. The discussion develops two interpretations of the documented patterns and reflects on what the methodological choices behind the analysis allow these interpretations to establish.

8.1 The institutional dimension of the Norwegian case

The structural channels reviewed in Chapter 2, including terms-of-trade effects, fiscal flows from petroleum activity, and monetary-policy responses to oil-related macroeconomic conditions, suggest that an oil-exposed economy should display some degree of structural exchange-rate response to oil prices. The robust low-stress oil response documented in Chapters 6 and 7 is consistent with this expectation. Beyond these mechanisms, Norway's institutional framework includes features that may also play a role in shaping the krone's response to oil prices.

Oil companies operating on the Norwegian continental shelf primarily earn their revenues in foreign currency but are subject to a combined marginal tax rate of 78 per cent on petroleum profits, with tax payments due in Norwegian kroner (Norwegian Petroleum, 2026). To meet these tax obligations, which constitute the main source of the government's petroleum revenues, companies convert foreign-currency revenues into NOK in the foreign exchange market. The government also receives substantial NOK revenues through dividends and share buybacks from its holding in Equinor, while its direct financial interest in oil and gas fields generates additional revenues primarily in foreign currency (Norges Bank, 2023b).

The state's net cash flow from petroleum activities is transferred to the Government Pension Fund Global, with transfers back to the central government budget guided by the fiscal rule (Holden, 2013), under which the structural non-oil budget deficit shall over time correspond to the expected real rate of return on the fund, currently set at 3 per cent, reduced from 4 per cent in 2017 (Ministry of Finance, 2017). Because the fund invests exclusively in foreign assets, most accumulated petroleum revenues are held abroad rather than entering the domestic economy directly, with Norges Bank conducting the foreign exchange transactions required to balance the government's NOK needs against transfers to and from the fund (Norges Bank, 2023a).

These institutional features may help explain why Norway's oil response appears stable across volatility regimes rather than activating primarily under stress. The petroleum-related flows operate through embedded channels that respond to oil prices in a continuous and predictable manner, rather than through market-driven mechanisms that may amplify or recede with sentiment and risk conditions. The institutional architecture therefore provides one possible explanation for the empirical pattern observed in Chapters 6 and 7.

8.2 The behavioural perspective and narrative activation

The empirical pattern documented in Section 7.8 stands in contrast with the regime structure for the non-Norway commodity groups in Chapter 6. Under the Brent specification, Norway is the only commodity-exposed group whose oil sensitivity against the diversified reference is similar in magnitude across market conditions, while the non-Norway groups converge toward Norway only under elevated volatility (*section 6.2.5*). Under the CTOT specification, the non-Norway groups show significant responses to their own commodity baskets in calm conditions, while the corresponding high-stress responses are no longer significant.

These observations raise two related questions about the role of oil in commodity-currency dynamics. The first is whether Norway's distinctness under Brent reflects a uniquely strong response to oil prices, or whether it partly reflects that oil constitutes a larger share of Norway's commodity exposure than is the case for the other groups. The fact that the non-Norway groups display similar low-volatility responses to their own commodity baskets under CTOT speaks to this question, although the analysis does not settle it. The second question is why the same non-Norway groups show tendencies toward stronger Brent sensitivity under stress, when their country-specific commodity exposures generate visible responses in calm conditions. Behavioural finance offers a lens on what the findings may reflect.

A familiar observation in behavioural finance is that decisions made under uncertainty rely on heuristics rather than full processing of the relevant information (Tversky & Kahneman, 1974), reflecting the broader influence of attention and availability biases on financial decision-making (Bachmann et al., 2018). Oil prices are unusually prominent in this respect. They are quoted continuously, appear routinely in market commentary, and are embedded in widely used analytical frameworks. From this perspective, the broad classification of certain currencies as "oil currencies" can function as a heuristic that organises a complex set of relationships into a simpler narrative (Shiller, 2017). One reading of the Brent regime-conditional pattern for non-Norway groups is that these heuristics gain influence specifically when uncertainty is elevated and market participants lean more heavily on salient classifications. This is offered as a lens for interpreting the pattern, not as a claim about what causes it.

A separate but related question is why such patterns would persist if some market participants recognise that the heuristic overstates the underlying relationship. Behavioural finance offers a familiar answer. Arbitrage has limits. Funding constraints and short investment horizons reduce the capacity of sophisticated traders to push prices back toward fundamentals (Shleifer & Vishny, 1997). Brunnermeier and Pedersen (2009) show that funding liquidity tightens precisely when prices move most aggressively, so deviations from fundamentals can be amplified instead of corrected in periods of stress. To the extent that the Brent regime-conditional pattern reflects a narrative-driven classification, these conditions could explain why such a pattern would emerge specifically in the same periods where arbitrage is most constrained.

What the analysis cannot determine is whether any specific group's response reflects fundamental forces, behavioural classification, or some mix of the two. Norway's stable oil response across volatility regimes is consistent with the institutional anchoring described in Section 8.1, but the analysis does not establish that institutional channels fully account for the response. The non-Norway groups show patterns that fit a narrative-driven reading under stress, but the country-specific commodity responses observed in calm conditions also have underlying drivers the analysis does not isolate. The behavioural perspective developed here outlines possible mechanisms that may underlie the observed patterns. Which mechanisms are actually at work for any given currency remains an open question that future research combining structural identification with behavioural lenses could address.

8.3 Methodological scope and structural limitations

The analysis rests on a sequence of methodological choices that have shaped what the empirical design can and cannot establish. With the empirical results in hand, these choices can now be assessed against the findings. The quarterly panel design implies a trade-off between capturing slower-moving fundamental drivers and tracking short-run movements. The choice to focus on the former rests on the assumption that lower-frequency observations filter out short-lived market noise to isolate the slower dynamics through which fundamentals are expected to operate, consistent with the lower-frequency focus adopted by Bjørnland et al. (2024).

The monthly robustness check in Section 7.1 illustrates this trade-off in concrete terms. The monthly specification produces a lower within R^2 than the quarterly baseline, consistent with higher-frequency variation introducing more noise relative to the fundamental signal the model is designed to capture. At the same time, the regime-conditional oil sensitivities are estimated more sharply at monthly frequency, reflecting the additional identifying variation in VIX regime transitions available at higher sampling rates. Both effects are direct expressions of the trade-off, and the qualitative pattern of the quarterly results is preserved across the two frequencies. It should be noted that within R^2 measures explanatory power for temporal variation around country-specific means after fixed effects have absorbed cross-country level differences, and is therefore systematically lower than the corresponding pooled R^2 would be. Together, these patterns confirm that the quarterly frequency was an appropriate choice for capturing fundamental drivers, while the monthly specification identifies the regime-conditional structure more sharply. The central qualitative conclusions hold across the two sampling rates.

Beyond the frequency dimension, the empirical design is also shaped by the choice of which countries to include in the panel. Several major oil-producing countries are not represented in the sample because they lack consistent quarterly series over the full sample period, do not operate freely floating exchange rate regimes, or both. Russia, Saudi Arabia, the United Arab Emirates, and Iran are notable examples. The sample therefore deliberately balances broad cross-country coverage with consistent data quality, where commodity-exporting economies with incomplete or poor quality data have been excluded to ensure comparability across all panel units.

A final dimension of the design concerns the empirical approach itself. Given the comparative and state-dependent questions this thesis addresses, the reduced-form panel design is well suited to the analysis. It is worth mentioning that the interpretations developed in Sections 8.1 and 8.2 extend beyond what this design can establish directly, since they discuss institutional and behavioural mechanisms as possible explanations for the observed patterns. Future research could complement the present analysis by applying structural identification strategies, such as the oil-market shock decomposition developed by Kilian (2009), to address which types of oil-price movements drive the differential responses documented here. The regime-conditional and group-level patterns documented in the present thesis would provide concrete hypotheses for such structural analysis to test, including whether Norway's stable response and the activation pattern observed for other commodity groups reflect different sensitivities to specific types of oil-market shocks.

9 Conclusion

This thesis has examined the role of oil prices in shaping movements in the Norwegian krone, asking whether oil contributes incremental explanatory power beyond standard macroeconomic fundamentals, whether the krone responds differently from other commodity-exposed currencies, and whether these relationships vary with the prevailing volatility regime. The analysis was conducted within a quarterly panel of fourteen economies covering 2000Q1 to 2025Q4, with Norway treated separately from non-Norwegian oil exporters, other commodity exporters, safe-haven currencies, and a diversified reference group. A sequence of three nested panel specifications allowed the role of oil to be assessed first against the macroeconomic baseline, then through heterogeneous cross-country exposure, and finally through regime-conditional interactions with the VIX.

The empirical evidence provides clear answers to the three research questions. Oil prices contribute meaningful incremental explanatory power for exchange-rate movements beyond the macroeconomic fundamentals. The Norwegian krone appreciates against the dollar when oil prices rise, and this appreciation is statistically distinguishable from the response of the diversified reference group, though not from the response of other oil-exporting currencies when averaged across the full sample. The relationship between oil prices and exchange rates varies systematically with the volatility regime, with the non-Norway commodity groups acquiring substantial oil sensitivity primarily under elevated volatility, while Norway's response is of comparable magnitude across both calm and stressed conditions.

These findings refine the conventional oil-currency narrative for the Norwegian krone. The krone is distinct within the commodity-currency class, but its distinctness lies in the structure of its oil response rather than in its average magnitude. The Norwegian krone is the only currency exhibiting similar magnitudes of oil sensitivity across market conditions relative to the diversified reference. Under elevated volatility, the other commodity-exposed currencies converge toward Norway's magnitude, and the cross-currency distinction narrows. The robustness analyses confirm that Norway's low-stress response is the most stably identified feature of the relationship, holding across alternative frequencies, sample periods, volatility thresholds, country compositions, crisis episodes, activity measures, and alternative measures of forward-looking uncertainty.

What underlies these patterns is a separate question. The regime-conditional behaviour of the other commodity-exposed currencies fits a behavioural reading in which narrative classifications gain force under stress, but the present analysis cannot establish whether this is what drives the response. Given the well-documented empirical difficulty of modelling exchange rates with standard macroeconomic fundamentals, an analysis that explicitly links behavioural biases to systematic drivers of exchange rates is a particularly interesting direction. The comparative and state-dependent patterns documented here point to commodity-exposed currencies and stress regimes as natural settings for examining such mechanisms.

References (APA 7)

Ahir, H., Bloom, N., & Furceri, D. (2022). *The world uncertainty index* (NBER Working Paper No. 29763). National Bureau of Economic Research. https://doi.org/10.3386/w29763

Akram, Q. F. (2004). Oil prices and exchange rates: Norwegian evidence. *The Econometrics Journal*, *7*(2), 476–504. https://doi.org/10.1111/j.1368-423X.2004.00140.x

Amano, R. A., & van Norden, S. (1998a). Oil prices and the rise and fall of the US real exchange rate. *Journal of International Money and Finance*, *17*(2), 299–316. https://doi.org/10.1016/S0261-5606(98)00004-7

Amano, R. A., & van Norden, S. (1998b). Exchange rates and oil prices. *Review of International Economics*, *6*(4), 683–694. https://doi.org/10.1111/1467-9396.00136

Attílio, L. A., & Mollick, A. V. (2026). Oil price shocks and interest rates in major economies. *International Economics and Economic Policy*, *23*(1), 14. https://doi.org/10.1007/s10368-025-00709-0

Bachmann, K. K., De Giorgi, E. G., & Hens, T. (2018). *Behavioral finance for private banking: From the art of advice to the science of advice* (2nd ed.). Wiley.

Baillie, R. T., Kapetanios, G., & Kim, K. H. (2026). Yes! Uncovered interest parity does hold in the long run. *Journal of International Money and Finance*, *160*, Article 103455. https://doi.org/10.1016/j.jimonfin.2025.103455

Balassa, B. (1964). The purchasing-power parity doctrine: A reappraisal. *Journal of Political Economy*, *72*(6), 584–596. https://doi.org/10.1086/258965

Baumeister, C., & Kilian, L. (2016). Forty years of oil price fluctuations: Why the price of oil may still surprise us. *Journal of Economic Perspectives*, *30*(1), 139–160. https://doi.org/10.1257/jep.30.1.139

Beckmann, J., Czudaj, R. L., & Arora, V. (2020). The relationship between oil prices and exchange rates: Revisiting theory and evidence. *Energy Economics*, *88*, Article 104772. https://doi.org/10.1016/j.eneco.2020.104772

Bergé, L. R., Butts, K., & McDermott, G. (2026). *Fast and user-friendly econometrics estimations: The R package* fixest (arXiv:2601.21749). arXiv https://arxiv.org/abs/2601.21749

Bjørnland, H. C., & Hungnes, H. (2002). *Fundamental determinants of the long run real exchange rate: The case of Norway* (Discussion Papers No. 326). Statistics Norway, Research Department. https://www.ssb.no/forskning/discussion-papers/fundamental-determinants-of-the-long-run-real-exchange-rate

Bjørnland, H. C., Brubakk, L., & Maffei-Faccioli, N. (2024). *Piecing the puzzle: Real exchange rates and long run fundamentals* (Norges Bank Working Paper No. 21/2024). Norges Bank. https://www.norges-bank.no/en/news-events/publications/Working-Papers/2024/wp-212024/

Bodart, V., Candelon, B., & Carpantier, J.-F. (2012). Real exchanges rates in commodity producing countries: A reappraisal. *Journal of International Money and Finance*, *31*(6), 1482–1502. https://doi.org/10.1016/j.jimonfin.2012.02.012

Brunnermeier, M. K., Nagel, S., & Pedersen, L. H. (2008). Carry trades and currency crashes. *NBER Macroeconomics Annual*, *23*, 313–347. https://doi.org/10.1086/593088

Brunnermeier, M. K., & Pedersen, L. H. (2009). Market liquidity and funding liquidity. *The Review of Financial Studies*, *22*(6), 2201–2238. https://doi.org/10.1093/rfs/hhn098

Cameron, A. C., & Miller, D. L. (2015). A practitioner's guide to cluster-robust inference. *Journal of Human Resources*, *50*(2), 317–372. https://doi.org/10.3368/jhr.50.2.317

Cashin, P., Céspedes, L. F., & Sahay, R. (2004). Commodity currencies and the real exchange rate. *Journal of Development Economics*, *75*(1), 239–268. https://doi.org/10.1016/j.jdeveco.2003.08.005

Chen, Y.-C., & Rogoff, K. (2003). Commodity currencies. *Journal of International Economics*, *60*(1), 133–160. https://doi.org/10.1016/S0022-1996(02)00072-7

Corden, W. M., & Neary, J. P. (1982). Booming sector and de-industrialisation in a small open economy. *The Economic Journal, 92*(368), 825–848. https://doi.org/10.2307/2232670

ter Ellen, S. (2016). *Nonlinearities in the relationship between oil price changes and movements in the Norwegian krone* (Staff Memo No. 18/2016). Norges Bank. https://www.norges-bank.no/en/news-events/publications/Staff-Memo/2016/Staff-Memo-182016/

ter Ellen, S., & Martinsen, K. (2016). *Oil price shocks and the Norwegian effective exchange rate – an SVAR approach* (Staff Memo No. 16/2016). Norges Bank. https://www.norges-bank.no/contentassets/d0d21c7e8c8c4ae5b619569a435a8c63/staff-memo-16_2016.pdf

Engel, C., & West, K. D. (2005). Exchange rates and fundamentals. *Journal of Political Economy*, *113*(3), 485–517. https://doi.org/10.1086/429137

Fama, E. F. (1984). Forward and spot exchange rates. *Journal of Monetary Economics*, *14*(3), 319–338. https://doi.org/10.1016/0304-3932(84)90046-1

Ferraro, D., Rogoff, K., & Rossi, B. (2015). Can oil prices forecast exchange rates? An empirical analysis of the relationship between commodity prices and exchange rates. *Journal of International Money and Finance*, *54*, 116–141. https://doi.org/10.1016/j.jimonfin.2015.03.001

Forbes, K. J., & Warnock, F. E. (2012). Capital flow waves: Surges, stops, flight, and retrenchment. *Journal of International Economics*, *88*(2), 235–251. https://doi.org/10.1016/j.jinteco.2012.03.006

Golub, S. S. (1983). Oil prices and exchange rates. *The Economic Journal*, *93*(371), 576–593. https://doi.org/10.2307/2232396

Gruss, B., & Kebhaj, S. (2019). *Commodity terms of trade: A new database* (Working Paper No. WP/19/21). International Monetary Fund. https://www.imf.org/en/Publications/WP/Issues/2019/01/24/Commodity-Terms-of-Trade-A-New-Database-46522

Habib, M. M., & Stracca, L. (2011). *Getting beyond carry trade: What makes a safe haven currency?* (ECB Working Paper Series No. 1288). European Central Bank. https://www.ecb.europa.eu/pub/pdf/scpwps/ecbwp1288.pdf

Holden, S. (2013). Avoiding the resource curse: The case of Norway. *Energy Policy*, *63*, 870–876. https://doi.org/10.1016/j.enpol.2013.09.010

Kilian, L. (2009). Not all oil price shocks are alike: Disentangling demand and supply shocks in the crude oil market. *American Economic Review*, *99*(3), 1053–1069. https://doi.org/10.1257/aer.99.3.1053

Kilian, L., & Lewis, L. T. (2011). Does the Fed respond to oil price shocks? *The Economic Journal*, *121*(555), 1047–1072. https://doi.org/10.1111/j.1468-0297.2011.02437.x

Kilian, L., & Zhou, X. (2022). Oil prices, exchange rates and interest rates. *Journal of International Money and Finance*, *126*, Article 102679. https://doi.org/10.1016/j.jimonfin.2022.102679

Krugman, P. (1983). Oil and the dollar. In J. S. Bhandari & B. H. Putnam (Eds.), *Economic interdependence and flexible exchange rates* (pp. 179–190). MIT Press.

Kutner, M. H., Nachtsheim, C. J., Neter, J., & Li, W. (2005). *Applied linear statistical models* (5th ed.). McGraw-Hill/Irwin.

Lustig, H., Roussanov, N., & Verdelhan, A. (2011). Common risk factors in currency markets. *The Review of Financial Studies*, *24*(11), 3731–3777. https://doi.org/10.1093/rfs/hhr068

Meese, R. A., & Rogoff, K. (1983). Empirical exchange rate models of the seventies: Do they fit out of sample? *Journal of International Economics*, *14*(1–2), 3–24. https://doi.org/10.1016/0022-1996(83)90017-X

Menkhoff, L., Sarno, L., Schmeling, M., & Schrimpf, A. (2012). Carry trades and global foreign exchange volatility. *The Journal of Finance*, *67*(2), 681–718. https://doi.org/10.1111/j.1540-6261.2012.01728.x

Newey, W. K., & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, *55*(3), 703–708. https://doi.org/10.2307/1913610

Norwegian Ministry of Finance. (2017). *Perspektivmeldingen 2017* [Long-term perspectives on the Norwegian economy 2017] (Meld. St. 29, 2016–2017). https://www.regjeringen.no/no/dokumenter/meld.-st.-29-20162017/id2546674/

Norges Bank. (2022). *Monetary policy report 1/2022*. https://www.norges-bank.no/en/news-events/publications/Monetary-Policy-Report/2022/mpr-12022/content/

Norges Bank. (2023a). *Foreign exchange transactions on behalf of the government*. https://www.norges-bank.no/en/topics/liquidity-and-markets/Foreign-exchange-purchases-for-GPFG/

Norges Bank. (2023b). *Norges Bank's sales of Norwegian kroner (NOK) in the FX market*. https://www.norges-bank.no/en/news-events/news/Articles-and-opinion-pieces/2023-05-23-sales-of-nok/

Norges Bank. (2025). *Monetary policy report 2/2025*. https://www.norges-bank.no/en/news-events/publications/Monetary-Policy-Report/2025/mpr-22025/web-report-mpr-22025/

Norges Bank. (2026). *Monetary policy report 1/2026*. https://www.norges-bank.no/en/news-events/publications/Monetary-Policy-Report/2026/mpr-12026/web-report-mpr-12026/

Norwegian Offshore Directorate & Ministry of Energy. (2026). *The petroleum tax system*. Retrieved May 22, 2026, from https://www.norskpetroleum.no/en/economy/petroleum-tax/

Obstfeld, M., & Rogoff, K. (1996). *Foundations of international macroeconomics*. MIT Press.

Obstfeld, M., & Rogoff, K. (2001). The six major puzzles in international macroeconomics: Is there a common cause? *NBER Macroeconomics Annual*, *15*, 339–390. https://doi.org/10.1086/654423

Pesaran, M. H., & Smith, R. (1995). Estimating long-run relationships from dynamic heterogeneous panels. *Journal of Econometrics*, *68*(1), 79–113. https://doi.org/10.1016/0304-4076(94)01644-F

Ranaldo, A., & Söderlind, P. (2010). Safe haven currencies. *Review of Finance*, *14*(3), 385–407. https://doi.org/10.1093/rof/rfq007

Rey, H. (2015). *Dilemma not trilemma: The global financial cycle and monetary policy independence* (NBER Working Paper No. 21162). National Bureau of Economic Research. https://doi.org/10.3386/w21162

Rogoff, K. (1996). The purchasing power parity puzzle. *Journal of Economic Literature*, *34*(2), 647–668. https://www.jstor.org/stable/2729217

Samuelson, P. A. (1964). Theoretical notes on trade problems. *The Review of Economics and Statistics*, *46*(2), 145–154. https://doi.org/10.2307/1928178

Shiller, R. J. (2017). Narrative economics. *American Economic Review*, *107*(4), 967–1004. https://doi.org/10.1257/aer.107.4.967

Shleifer, A., & Vishny, R. W. (1997). The limits of arbitrage. *The Journal of Finance*, *52*(1), 35–55. https://doi.org/10.1111/j.1540-6261.1997.tb03807.x

Tversky, A., & Kahneman, D. (1974). Judgment under uncertainty: Heuristics and biases. *Science*, *185*(4157), 1124–1131. https://doi.org/10.1126/science.185.4157.1124

Wooldridge, J. M. (2010). *Econometric analysis of cross section and panel data* (2nd ed.). MIT Press.

AI Declaration

Declaration on the use of AI tools in the work on this master’s thesis

Name (and version) of the AI tool: Claude Opus 4.7

Purpose of using the tool: Research, source finding, discussing relevant topics, and assessing and improving precision in language.

Name (and version) of the AI tool: Claude Code Opus 4.7

Purpose of using the tool: General coding assistance

Name (and version) of the AI tool: ChatGPT-5.5

Purpose of using the tool: Proofreading and language precision

Name (and version) of the AI tool: Grammarly

Purpose of using the tool: General spelling and grammar

Name (and version) of the AI tool: Word (in-built spelling software)

Purpose of using the tool: General spelling and grammar

We are aware that we are responsible for all content of this master's thesis, including the parts where AI tools are used. We are responsible for ensuring that the thesis complies with ethical rules for privacy and publication.

Appendix

A Residual diagnostics:

A.1 Tables

Table A.1: Correlation matrix

Variable	FX	Interest rate	Inflation	GDP	Oil price	Vix	Norway	Oil country	Commo dity	Safe haven	High Vix
FX	1.000	0.047	0.317	-0.079	-0.295	0.225	-0.003	0.036	0.008	-0.029	0.120
Interest rate		1.000	0.468	0.033	-0.024	0.155	-0.078	0.490	0.134	-0.404	0.101
Inflation			1.000	-0.019	-0.235	0.241	-0.047	0.258	0.099	-0.285	0.110
GDP				1.000	0.238	0.024	-0.018	0.030	0.072	-0.066	0.060
Oil Price					1.000	-0.284	0.001	0.002	-0.002	0.001	-0.131
Vix						1.000	0.000	0.000	0.000	0.000	0.772
Norway							1.000	-0.177	-0.175	-0.114	0.000
Oil country								1.000	-0.399	-0.260	0.001
Commodity									1.000	-0.257	-0.001
Safe Haven										1.000	0.001
High Vix											1.000

Note. Pairwise Pearson correlations across all country-quarter observations in the full panel (N = 1,434).

Table A.2: Volatility Inflation Index

Regressor	GVIF	Df
Macro fundamentals		
Interest-rate differential	5.753	1
Inflation differential	2.433	1
GDP-growth differential	1.550	1
Oil interactions		
Other oil × oil (θ_O)	6.180	1
Other commodity × oil (θ_C)	6.062	1
Safe haven × oil (θ_S)	4.350	1
Norway × oil (θ_N)	3.472	1
Volatility regime interactions		
Other oil × HighVIX (γ_O)	2.967	1
Other commodity × HighVIX (γ_C)	2.943	1
Safe haven × HighVIX (γ_S)	2.207	1
Norway × HighVIX (γ_N)	1.793	1
Triple interactions		
Other oil × oil × HighVIX (φ_O)	6.080	1
Other commodity × oil × HighVIX (φ_C)	6.039	1
Safe haven × oil × HighVIX (φ_S)	4.350	1
Norway × oil × HighVIX (φ_N)	3.516	1

Note. Generalised variance inflation factors for the regressors in model 3.

A.2 Figures

Figure A.1: Average absolute residuals over time, GFC and Covid quarters shaded

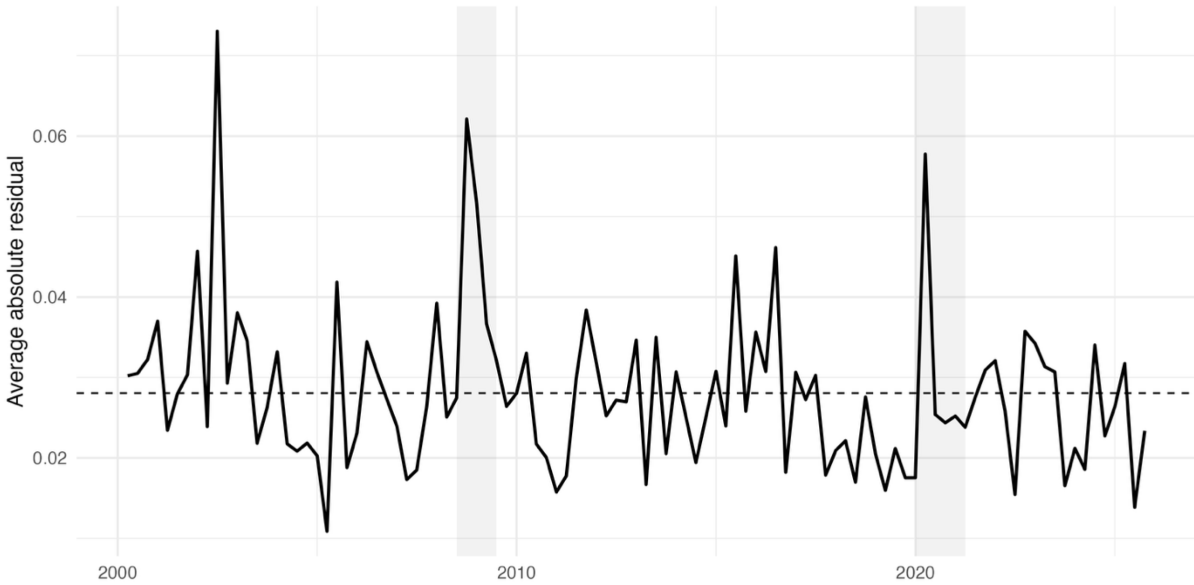


Figure A.2: Boxplots for each country, colours based on commodity group

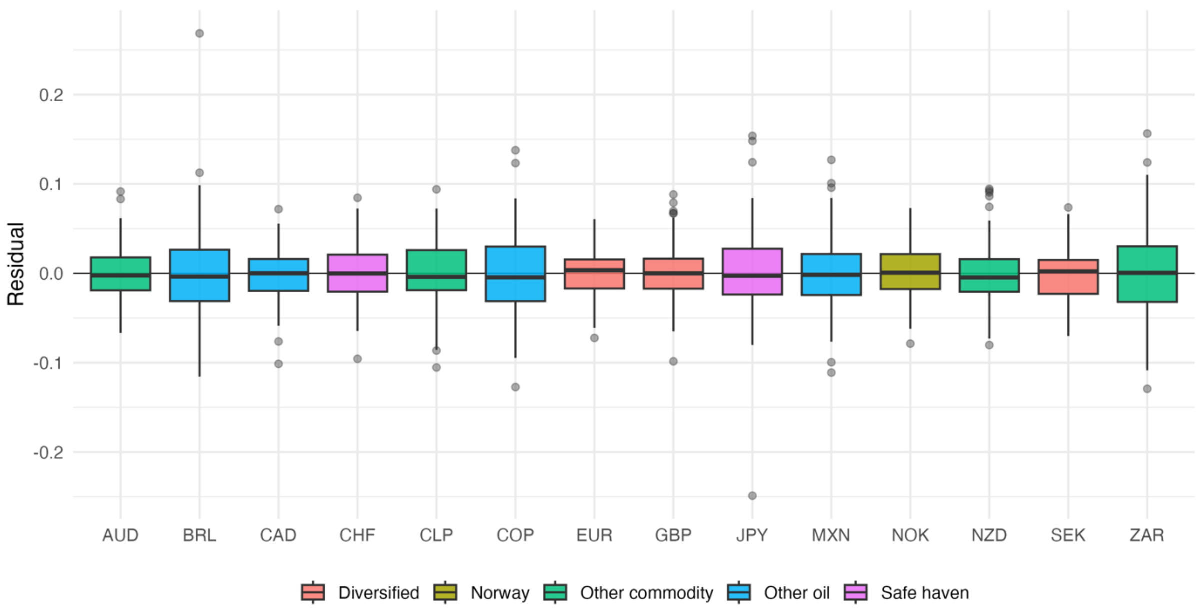


Figure A.3: Residuals VS. Fitted Values

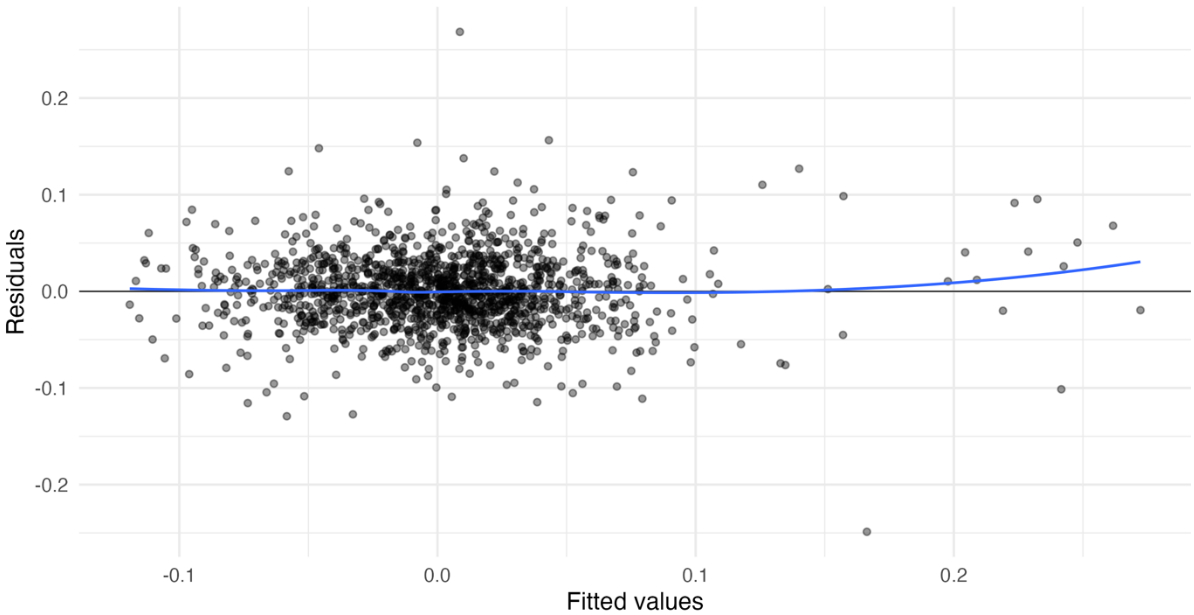


Figure A.4: Residual distribution in histogram

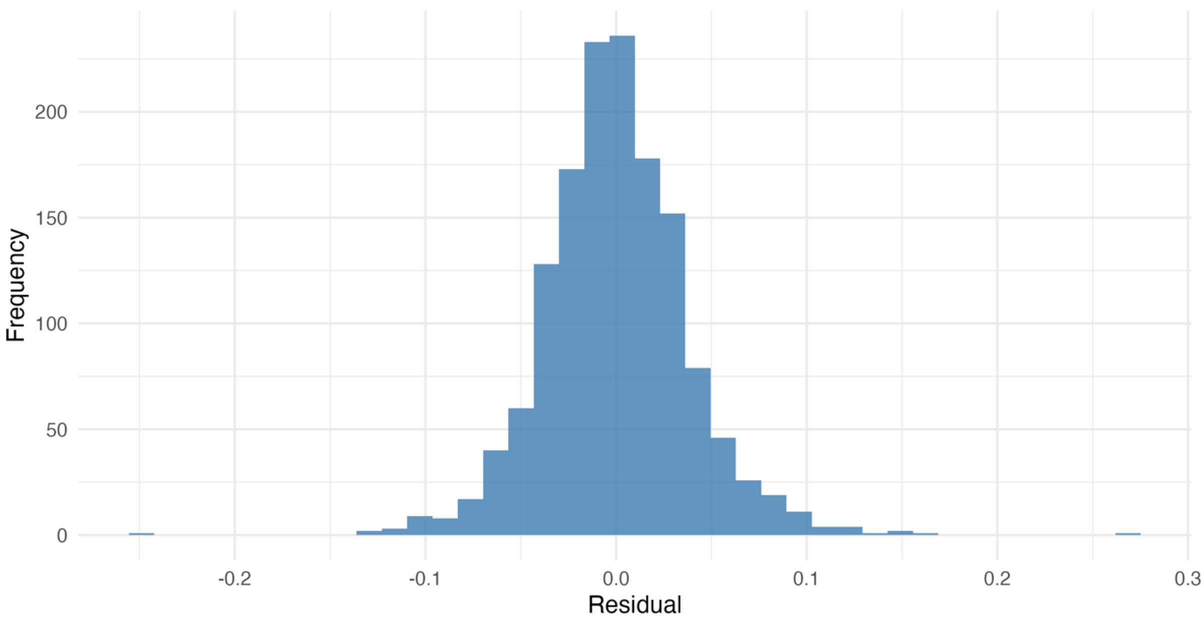
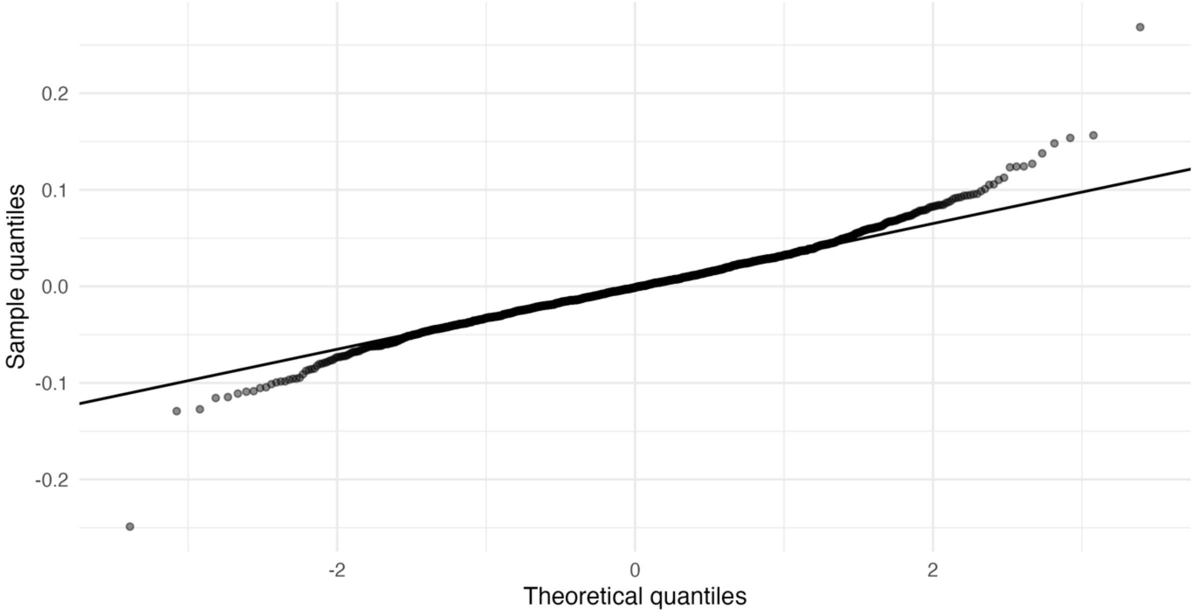


Figure A.5: Residual distribution in qq-plot



B Robustness Regressions

Table B.1: Robustness 7.2: VIX p50 threshold

	Model 1	Model 2	Model 3
GDP-growth differential	-0.1641	-0.0993	-0.0806
	(0.1626)	(0.1378)	(0.1327)
Inflation differential	0.9985***	0.9699***	0.9940***
	(0.2513)	(0.2586)	(0.2740)
Interest-rate differential	-0.0031***	-0.0032***	-0.0033***
	(0.0008)	(0.0008)	(0.0008)
Norway × oil		-0.0459***	-0.0444***
		(0.0098)	(0.0127)
Other commodity × oil		-0.0341	0.0093
		(0.0200)	(0.0154)
Other oil × oil		-0.0393**	0.0177
		(0.0154)	(0.0246)
Safe haven × oil		0.0678**	0.0293*
		(0.0246)	(0.0148)
Norway × HighVIX (p50)			-0.0002
			(0.0022)
Other commodity × HighVIX (p50)			0.0009
			(0.0033)
Other oil × HighVIX (p50)			0.0068
			(0.0046)
Safe haven × HighVIX (p50)			-0.0087**
			(0.0036)
Norway × oil × HighVIX (p50)			-0.0014
			(0.0308)
Other commodity × oil × HighVIX (p50)			-0.0618
			(0.0407)
Other oil × oil × HighVIX (p50)			-0.0816*
			(0.0454)
Safe haven × oil × HighVIX (p50)			0.0557
			(0.0376)
Observations	1434	1434	1434
Within R^2	0.032	0.073	0.091

*Note. Full output for the p50 robustness check reported in Table 7.2. Country clustered standard errors. Significance levels: : *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.*

Table B.2: Robustness 7.2: VIX p90 threshold

	Model 1	**Model 2**	**Model 3**
GDP-growth differential	-0.1641	-0.0993	-0.0605
	(0.1626)	(0.1378)	(0.1221)
Inflation differential	0.9985***	0.9699***	0.9164***
	(0.2513)	(0.2586)	(0.2344)
Interest-rate differential	-0.0031***	-0.0032***	-0.0029***
	(0.0008)	(0.0008)	(0.0008)
Norway × oil		-0.0459***	-0.0457***
		(0.0098)	(0.0023)
Other commodity × oil		-0.0341	-0.0127
		(0.0200)	(0.0130)
Other oil × oil		-0.0393**	-0.0039
		(0.0154)	(0.0127)
Safe haven × oil		0.0678**	0.0317***
		(0.0246)	(0.0073)
Norway × HighVIX (p90)			-0.0054**
			(0.0024)
Other commodity × HighVIX (p90)			0.0042
			(0.0111)
Other oil × HighVIX (p90)			0.0460***
			(0.0130)
Safe haven × HighVIX (p90)			-0.0111***
			(0.0031)
Norway × oil × HighVIX (p90)			-0.0034
			(0.0222)
Other commodity × oil × HighVIX (p90)			-0.0436
			(0.0384)
Other oil × oil × HighVIX (p90)			-0.0347
			(0.0354)
Safe haven × oil × HighVIX (p90)			0.0726
			(0.0439)
Observations	1434	1434	1434
Within R^2	0.032	0.073	0.114

*Note. Full output for the p90 robustness check reported in Table 7.2. Country clustered standard errors. Significance levels: : *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.*

Table B.3: Robustness 7.3: Brazil excluded

	Model 1	**Model 2**	**Model 3**
GDP-growth differential	-0.1760	-0.1285	-0.0859
	(0.1782)	(0.1505)	(0.1402)
Inflation differential	0.8536***	0.8468***	0.8595***
	(0.2600)	(0.2723)	(0.2665)
Interest-rate differential	-0.0030**	-0.0031**	-0.0032**
	(0.0012)	(0.0012)	(0.0013)
Norway × oil		-0.0472***	-0.0488***
		(0.0100)	(0.0083)
Other commodity × oil		-0.0342	0.0036
		(0.0201)	(0.0161)
Other oil × oil		-0.0269**	0.0246
		(0.0115)	(0.0206)
Safe haven × oil		0.0671**	0.0248*
		(0.0246)	(0.0130)
Norway × HighVIX			0.0024**
			(0.0010)
Other commodity × HighVIX			0.0034
			(0.0027)
Other oil × HighVIX			0.0094
			(0.0054)
Safe haven × HighVIX			-0.0103***
			(0.0010)
Norway × oil × HighVIX			0.0058
			(0.0286)
Other commodity × oil × HighVIX			-0.0604*
			(0.0311)
Other oil × oil × HighVIX			-0.0816
			(0.0509)
Safe haven × oil × HighVIX			0.0666*
			(0.0344)
Observations	1331	1331	1331
Within R^2	0.024	0.067	0.093

*Note. Full output for the exclude Brazil robustness check reported in Table 7.3. Country clustered standard errors. Significance levels: : *** p < 0.01, ** p < 0.05, * p < 0.10.*

Table B.4: Robustness 7.3: Mexico excluded

	Model 1	**Model 2**	**Model 3**
GDP-growth differential	-0.0419	0.0018	0.0187
	(0.1333)	(0.1131)	(0.1132)
Inflation differential	0.9891***	0.9818***	0.9948***
	(0.2986)	(0.3031)	(0.3068)
Interest-rate differential	-0.0027***	-0.0028***	-0.0028***
	(0.0008)	(0.0008)	(0.0008)
Norway × oil		-0.0426***	-0.0481***
		(0.0096)	(0.0086)
Other commodity × oil		-0.0333	0.0042
		(0.0206)	(0.0162)
Other oil × oil		-0.0363*	0.0059
		(0.0192)	(0.0164)
Safe haven × oil		0.0695**	0.0258*
		(0.0248)	(0.0140)
Norway × HighVIX			0.0023***
			(0.0007)
Other commodity × HighVIX			0.0029
			(0.0026)
Other oil × HighVIX			0.0146**
			(0.0065)
Safe haven × HighVIX			-0.0100***
			(0.0011)
Norway × oil × HighVIX			0.0112
			(0.0286)
Other commodity × oil × HighVIX			-0.0605*
			(0.0319)
Other oil × oil × HighVIX			-0.0632
			(0.0420)
Safe haven × oil × HighVIX			0.0684*
			(0.0337)
Observations	1331	1331	1331
Within R^2	0.026	0.069	0.093

*Note. Full output for the exclude Mexico robustness check reported in Table 7.3. Country clustered standard errors. Significance levels: : *** p < 0.01, ** p < 0.05, * p < 0.10.*

Table B.5: Robustness 7.4: Exclude COVID 2020Q1-2020Q4

	Model 1	**Model 2**	**Model 3**
GDP-growth differential	0.0944	0.0760	0.0332
	(0.1232)	(0.1147)	(0.1144)
Inflation differential	1.1713***	1.1402***	1.1065***
	(0.2553)	(0.2534)	(0.2656)
Interest-rate differential	-0.0027***	-0.0029***	-0.0030***
	(0.0009)	(0.0009)	(0.0009)
Norway × oil		-0.0276***	-0.0475***
		(0.0076)	(0.0086)
Other commodity × oil		-0.0178	0.0045
		(0.0167)	(0.0165)
Other oil × oil		-0.0104	0.0186
		(0.0123)	(0.0175)
Safe haven × oil		0.0797**	0.0264*
		(0.0287)	(0.0144)
Norway × HighVIX			0.0015
			(0.0015)
Other commodity × HighVIX			0.0039
			(0.0029)
Other oil × HighVIX			0.0097**
			(0.0044)
Safe haven × HighVIX			-0.0113***
			(0.0015)
Norway × oil × HighVIX			0.0419
			(0.0315)
Other commodity × oil × HighVIX			-0.0453
			(0.0400)
Other oil × oil × HighVIX			-0.0559
			(0.0398)
Safe haven × oil × HighVIX			0.1061**
			(0.0462)
Observations	1378	1378	1378
Within R^2	0.035	0.061	0.085

*Note. Full output for the exclude Covid robustness check reported in Table 7.4. Country clustered standard errors. Significance levels: : *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.*

Table B.6: Robustness 7.4: Exclude GFC 2008Q3-2009Q2

	Model 1	**Model 2**	**Model 3**
GDP-growth differential	-0.2028	-0.1537	-0.1034
	(0.1802)	(0.1461)	(0.1288)
Inflation differential	0.9187***	0.9122***	0.9975***
	(0.2683)	(0.2780)	(0.2812)
Interest-rate differential	-0.0035***	-0.0035***	-0.0036***
	(0.0007)	(0.0007)	(0.0007)
Norway × oil		-0.0599***	-0.0479***
		(0.0064)	(0.0085)
Other commodity × oil		-0.0362*	0.0042
		(0.0200)	(0.0157)
Other oil × oil		-0.0471***	0.0183
		(0.0127)	(0.0179)
Safe haven × oil		0.0295***	0.0256*
		(0.0084)	(0.0132)
Norway × HighVIX			0.0052***
			(0.0011)
Other commodity × HighVIX			0.0079
			(0.0051)
Other oil × HighVIX			0.0189***
			(0.0054)
Safe haven × HighVIX			-0.0046*
			(0.0024)
Norway × oil × HighVIX			-0.0188
			(0.0279)
Other commodity × oil × HighVIX			-0.0814**
			(0.0289)
Other oil × oil × HighVIX			-0.1299**
			(0.0432)
Safe haven × oil × HighVIX			0.0098
			(0.0262)
Observations	1379	1379	1379
Within R^2	0.037	0.058	0.085

*Note. Full output for the exclude GFC robustness check reported in Table 7.4. Country clustered standard errors. Significance levels: : *** p < 0.01, ** p < 0.05, * p < 0.10.*

Table B.7: Robustness 7.4: Exclude oil crash 2014Q4-2016Q1

	Model 1	**Model 2**	**Model 3**
GDP-growth differential	-0.1464	-0.0908	-0.0393
	(0.1679)	(0.1435)	(0.1354)
Inflation differential	0.9283***	0.9338***	0.9010**
	(0.2943)	(0.2992)	(0.3033)
Interest-rate differential	-0.0032***	-0.0033***	-0.0033***
	(0.0009)	(0.0008)	(0.0009)
Norway × oil		-0.0496***	-0.0601***
		(0.0112)	(0.0103)
Other commodity × oil		-0.0403*	-0.0002
		(0.0208)	(0.0164)
Other oil × oil		-0.0431**	0.0334
		(0.0184)	(0.0258)
Safe haven × oil		0.0641**	-0.0058
		(0.0286)	(0.0208)
Norway × HighVIX			0.0026***
			(0.0008)
Other commodity × HighVIX			0.0043*
			(0.0023)
Other oil × HighVIX			0.0165**
			(0.0065)
Safe haven × HighVIX			-0.0119***
			(0.0013)
Norway × oil × HighVIX			0.0196
			(0.0300)
Other commodity × oil × HighVIX			-0.0563
			(0.0336)
Other oil × oil × HighVIX			-0.1049**
			(0.0477)
Safe haven × oil × HighVIX			0.0986***
			(0.0315)
Observations	1350	1350	1350
Within R^2	0.030	0.071	0.106

*Note. Full output for the exclude oil crash robustness check reported in Table 7.4. Country clustered standard errors. Significance levels: : *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.*

Table B.8: Robustness 7.5: Norway mainland GDP

	Model 1	**Model 2**	**Model 3**
GDP-growth differential	-0.1964	-0.1243	-0.0882
	(0.1665)	(0.1462)	(0.1316)
Inflation differential	0.9907***	0.9652***	0.9941***
	(0.2532)	(0.2598)	(0.2599)
Interest-rate differential	-0.0031***	-0.0032***	-0.0032***
	(0.0008)	(0.0008)	(0.0008)
Norway × oil		-0.0462***	-0.0477***
		(0.0096)	(0.0085)
Other commodity × oil		-0.0343	0.0044
		(0.0200)	(0.0160)
Other oil × oil		-0.0391**	0.0183
		(0.0154)	(0.0174)
Safe haven × oil		0.0674**	0.0258*
		(0.0246)	(0.0134)
Norway × HighVIX			0.0025***
			(0.0007)
Other commodity × HighVIX			0.0033
			(0.0026)
Other oil × HighVIX			0.0130**
			(0.0051)
Safe haven × HighVIX			-0.0099***
			(0.0010)
Norway × oil × HighVIX			0.0054
			(0.0284)
Other commodity × oil × HighVIX			-0.0620*
			(0.0313)
Other oil × oil × HighVIX			-0.0892*
			(0.0422)
Safe haven × oil × HighVIX			0.0656*
			(0.0341)
Observations	1434	1434	1434
Within R^2	0.032	0.073	0.100

*Note. Full output for the robustness check discussed in Section 7.5. Country-clustered standard errors. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.*

Table B.9: Robustness 7.6: WUI replaces VIX in regime classification

	Model 1	**Model 2**	**Model 3**
GDP-growth differential	-0.1641	-0.0993	-0.1246
	(0.1626)	(0.1378)	(0.1358)
Inflation differential	0.9985***	0.9699***	0.9793***
	(0.2513)	(0.2586)	(0.2573)
Interest-rate differential	-0.0031***	-0.0032***	-0.0032***
	(0.0008)	(0.0008)	(0.0008)
Norway × oil		-0.0459***	-0.0280**
		(0.0098)	(0.0106)
Other commodity × oil		-0.0341	-0.0332
		(0.0200)	(0.0249)
Other oil × oil		-0.0393**	-0.0218
		(0.0154)	(0.0198)
Safe haven × oil		0.0678**	0.0843**
		(0.0246)	(0.0390)
Norway × HighWUI			0.0045*
			(0.0024)
Other commodity × HighWUI			0.0041
			(0.0037)
Other oil × HighWUI			-0.0012
			(0.0054)
Safe haven × HighWUI			0.0113
			(0.0108)
Norway × oil × HighWUI			-0.0417***
			(0.0125)
Other commodity × oil × HighWUI			-0.0015
			(0.0154)
Other oil × oil × HighWUI			-0.0400**
			(0.0177)
Safe haven × oil × HighWUI			-0.0358
			(0.0335)
Observations	1434	1434	1434
Within R^2	0.032	0.073	0.078

*Note. Full output for WUI robustness check reported in Table 7.6. Country clustered standard errors. Significance levels: : *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.*

Table B.10: Robustness 7.7: CTOT replaces oil

	Model 1	**Model 2**	**Model 3**
GDP-growth differential	-0.1641	-0.1849	-0.1822
	(0.1626)	(0.1590)	(0.1586)
Inflation differential	0.9985***	0.9258***	0.8871***
	(0.2513)	(0.2736)	(0.2736)
Interest-rate differential	-0.0031***	-0.0032***	-0.0032***
	(0.0008)	(0.0007)	(0.0007)
Norway × CTOT		-0.4713***	-0.6554***
		(0.1458)	(0.0783)
Other commodity × CTOT		-0.5546**	-1.0058***
		(0.2435)	(0.1440)
Other oil × CTOT		-1.3123	-1.7112***
		(0.7719)	(0.4212)
Norway × HighVIX			0.0063**
			(0.0028)
Other commodity × HighVIX			0.0109**
			(0.0042)
Other oil × HighVIX			0.0203***
			(0.0065)
Norway × CTOT × HighVIX			0.3438*
			(0.1613)
Other commodity × CTOT × HighVIX			0.8745
			(0.5104)
Other oil × CTOT × HighVIX			1.1652
			(0.9424)
Observations	1434	1434	1434
Within R^2	0.032	0.049	0.061

*Note. Full output for the CTOT robustness check reported in Table 7.7. Safe haven is included in the baseline due to low commodity activity. Country clustered standard errors. Significance levels: : *** p < 0.01, ** p < 0.05, * p < 0.10.*

Table B.11: Regime conditional contrasts at monthly frequency

Regime	Equation	Restriction	Estimate	*p*-value
Low VIX	Eq. 3.4	$\theta_N - \theta_O = 0$	−0.0361***	0.0012
Low VIX	Eq. 3.4	$\theta_N - \theta_C = 0$	−0.0373***	< 0.001
High VIX	Eq. 3.4	$(\theta_N + \varphi_N) - (\theta_O + \varphi_O) = 0$	−0.0004	0.9772
High VIX	Eq. 3.4	$(\theta_N + \varphi_N) - (\theta_C + \varphi_C) = 0$	+0.0014	0.7972

Note. Low VIX tests are one-sided, High VIX tests are two-sided. Cluster robust standard errors.